HOSPITALIZATION CHARACTERISTICS AND PREDICTORS OF RENAL FAILURE OF INPATIENTS IN THE UNITED STATES

By

Rexford Acheampong

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Department of Health Informatics
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School of Health Professions

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Final Dissertation Defense Approval Form

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By

Rexford Acheampong

Dissertation Committee:

Shankar Srinivasan, Ph.D., Committee Chair
Dinesh Mittal, Ph.D., Co-Chairperson
Riddhi Vyas, Ph.D., Committee Member

Approved by the Dissertation Committee:

_____ Date: _____

_____ Date: _____

_____ Date: _____

Abstract

The problem addressed was a renal failure and its association with age, gender, race, SES, number of procedures, insurance type, comorbidities, length of stay, admission type, risk of mortality, type of CKD, illness severity, number of chronic conditions, in a cohort of inpatients in the United States. The purpose of this study was to investigate the relationship between renal failure and length of stay, total charges, and their association with age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. This study's research methodology and design was a quantitative correlational design study. This study's data collection instrument was archival data of the selected inpatients in the United States. The researcher targeted the data from Nationwide Inpatient Sample (NIS) datasets for 2010, 2011, and 2012. The resulting quantitative data analysis was conducted using the statistical software suite Statistical Package for the Social Sciences (SPSS) version 23. Data analysis established a significant relationship between the length of stay, the total charges and renal failure. The logistic regression model was statistically significant, and the findings demonstrated that both length of stay and the total charge were significantly related to renal failure. Such a finding is highly plausible given renal failure may result in increased hospitalization and accordingly greater amount of care resulting in increased charges as well. The overall research though illuminative of significant predictors for kidney failure is however limited being confined to a retrospective study design and accordingly devoid of any possibilities of making causal inferences as would be possible in a randomized clinical trial (RCT). Future research should be conducted using different types of research design including a RCT, a qualitative research design and such to provide a more comprehensive view of how the length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities are all related to renal failure.

Keywords: renal failure, chronic kidney disease, Nationwide Inpatient Sample, length of stay, total charges, health disparities, binary logistic regression

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Dedication

I dedicate this dissertation work to my family. I love you deeply with all my heart. To my wife Joycelyn, you have been a listener and a supporter of all my endeavors. Your partnership, steadfastness, and love sustain me. To my son Preston, remember all things are possible. Never be afraid to pursue your dreams and goals. You are a precious gift from God. I love you without measure. Thank you for your patience as I pursued and completed this degree. To my parents, thank you for raising your children to think for themselves. I am grateful for my mom who always told me I could achieve anything if I put my mind to it. I also dedicate this work and give special thanks to Nana Kyemereh Kokoankra II of Wassa Dominase for your encouragement. Finally, I dedicate this thesis to God Almighty my creator, my strong pillar, my source of inspiration, wisdom, knowledge and understanding. He has been the source of my strength throughout this program and on His wings only have I soared.

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CHAPTER I: INTRODUCTION

Chronic kidney disease (CKD) can eventually cause renal failure and gradual loss of kidney function (Patel et al., 2017). Renal failure is defined as the kidneys' inability to perform excretory functions, resulting in the retention of nitrogenous waste products in the blood. Renal failure has a high morbidity and mortality rate, costing the healthcare system billions annually.

CKD brings significant dyslipidemia and lipid and lipoprotein metabolism changes, which may result in renal failure (Chen et al., 2019). CKD is defined as the presence of kidney damage or an estimated glomerular filtration rate (eGFR) less than 60 mL/min/1.73 m^2 that lasts for three months or longer, regardless of the cause (Vaidya & Aeddula, 2022). A progressive loss of kidney function eventually necessitates renal replacement therapy (dialysis or transplantation). Pathologic abnormalities in urinary sediment, urinary albumin excretion rates, or increased urinary albumin excretion rates are all examples of kidney damage. Cardiovascular issues and the progression of CKD may result from the associated dyslipidemia (Chen et al., 2019). In the US, 37 million adults, or 15% of the population, have CKD, but nearly 90% are unaware of their condition (Centers for Disease Control and Prevention [CDC], 2022).

This study focuses on renal failure and its association with length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. Chapter I will focus on the introduction. The researcher will introduce and discuss the background of the problem, the statement of the problem, the definitions, the hypotheses, and the need for the study.

Background of the Problem

Acute kidney failure occurs when the kidneys are suddenly unable to filter waste from the blood. Dangerous levels of waste may build up when kidney function declines, and the chemical composition of the blood may become unbalanced. Acute kidney injury, also known as acute renal failure, develops quickly, typically in less than a few days. The majority of cases of acute kidney failure occur in patients who are already hospitalized, particularly in critically ill patients

who require intensive care. Acute kidney failure is serious and needs aggressive care. Acute kidney failure, however, might be curable. An individual may regain normal or nearly normal kidney function if they are otherwise healthy. According to the Mayo Clinic (2022), signs and symptoms of acute kidney failure may include the following:

- Decreased urine output, although occasionally urine output remains normal.
- Fluid retention, causing swelling in your legs, ankles, or feet.
- Shortness of breath
- Fatigue
- Confusion
- Nausea
- Weakness
- Irregular heartbeat
- Chest pain or pressure
- Seizures or coma in severe cases

Acute kidney failure may occasionally go undetected and only be discovered through lab tests ordered for another reason. Acute kidney failure almost always occurs with another medical condition or event. Conditions that can increase the risk of acute kidney failure include:

- Being hospitalized
- Advanced age
- Blockages in the blood vessels in your arms or legs (peripheral artery disease)
- Diabetes
- High blood pressure
- Heart failure
- Kidney diseases
- Liver diseases
- Certain cancers and their treatments

To ensure patients' survival, treatment management necessitates dialysis or kidney transplantation. Dialysis is using a dialysis machine to cleanse the blood inside or outside the body; transplantation is receiving a live kidney from a donor (National Kidney Foundation, 2016). Social factors such as race, ethnicity, and age have been attributed to renal failure. Hypertension and diabetes are the two other major causes of chronic renal disease that lead to end-stage renal disease. Nephrologists attempt to educate their patients on managing their hypertension and glucose levels to delay the onset of chronic renal failure and prevent the progression to end-stage renal disease (National Kidney Foundation, 2016).

According to USRDS (2016), renal failure patients have a higher risk of hospitalization. The Centers for Medicare and Medicaid have restricted financial reimbursement for readmitted patients within 30 days of discharge, and this system keeps track of payments made to hospitals and hemodialysis centers. According to Hakim and Collins (2014), preventing unplanned hospitalization for hemodialysis patients could improve health outcomes and result in significant cost savings. According to Bharadwaj et al. (2016), palliative care may reduce hospitalization rates and length of stay.

Statement of the Problem

The problem to be addressed is renal failure and its association with age, gender, race, SES, number of procedures, insurance type, comorbidities, length of stay, admission type, risk of mortality, type of CKD, illness severity, number of chronic conditions, in a cohort of inpatients in the United States. In the United States, 15% of adults, or approximately 37 million people, have renal failure (CDC, 2022). Those with CKD made up 3.5% of adult visits to the emergency room in the United States between 2015 and 2016 (CDC, 2022). The specific issue is that among inpatients with CKD, it is still unclear how renal failure relates to factors such as length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities.

Definitions

Chronic kidney disease (CKD). Chronic kidney failure is a long-term kidney disease that leads to renal failure and a gradual loss of kidney function over time. (Patel et al., 2017).

Comorbidity. The medicine defines comorbidity as one or more additional conditions that frequently co-occur with a primary condition. (McQueen et al., 2017).

Socioeconomic status (SES). Socioeconomic status (SES) includes a person's income, level of education, financial security, subjective judgments of their social status and class, characteristics of their quality of life, and the opportunities and privileges available to them in society (Golestaneh et al., 2018).

Renal failure. In this study, renal failure is the Agency for Healthcare Research and Quality (AHRQ) comorbidity indicator for renal failure (CM_RENLFAIL), coded 1 when the condition is recorded on the discharge abstract and 0 otherwise (Table 1). It is the dependent variable for all ten research questions.

Hypotheses

The following research questions and hypotheses will be addressed in the study:

RQ1: Is there a relationship between renal failure and length of stay?
H_0: There is no significant relationship between renal failure and length of stay.
H_1: There is a significant relationship between renal failure and length of stay.

RQ2: Is there a relationship between renal failure and total charges?
H_0: There is no significant relationship between renal failure and total charges.
H_1: There is a significant relationship between renal failure and total charges.

RQ3: Is there a relationship between renal failure and age?
H_0: There is no significant relationship between renal failure and age.
H_1: There is a significant relationship between renal failure and age.

RQ4: Is there a relationship between renal failure and gender?
H_0: There is no significant relationship between renal failure and gender.
H_1: There is a significant relationship between renal failure and gender.

RQ5: Is there a relationship between renal failure and race?

H_0: There is no significant relationship between renal failure and race.

H_1: There is a significant relationship between renal failure and race.

RQ6: Is there a relationship between renal failure and socioeconomic status (SES)?

H_0: There is no significant relationship between renal failure and socioeconomic status (SES).

H_1: There is a significant relationship between renal failure and socioeconomic status (SES).

RQ7: Is there a relationship between renal failure and the number of procedures?

H_0: There is no significant relationship between renal failure and the number of procedures.

H_1: There is a significant relationship between renal failure and the number of procedures.

RQ8: Is there a relationship between renal failure and the number of diagnoses?

H_0: There is no significant relationship between renal failure and the number of diagnoses.

H_1: There is a significant relationship between renal failure and the number of diagnoses.

RQ9: Is there a relationship between renal failure and comorbidities?

H_0: There is no significant relationship between renal failure and comorbidities.

H_1: There is a significant relationship between renal failure and comorbidities.

RQ10: Do length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively predict renal failure?

H_0: Length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively do not significantly predict renal failure.

H_1: Length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively significantly predict renal failure.

Figure 1 summarizes how these research questions map onto the study variables and the analytic strategy.

The Need for the Study

This study may advance knowledge and research on renal failure and CKD. To comprehend the causes and clinical effects of racial differences in CKD, Meghani et al. (2020) suggested larger studies. In a cohort of inpatients in the US, associations between renal failure and length of stay, total costs, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities are still unknown, according to a review of the relevant

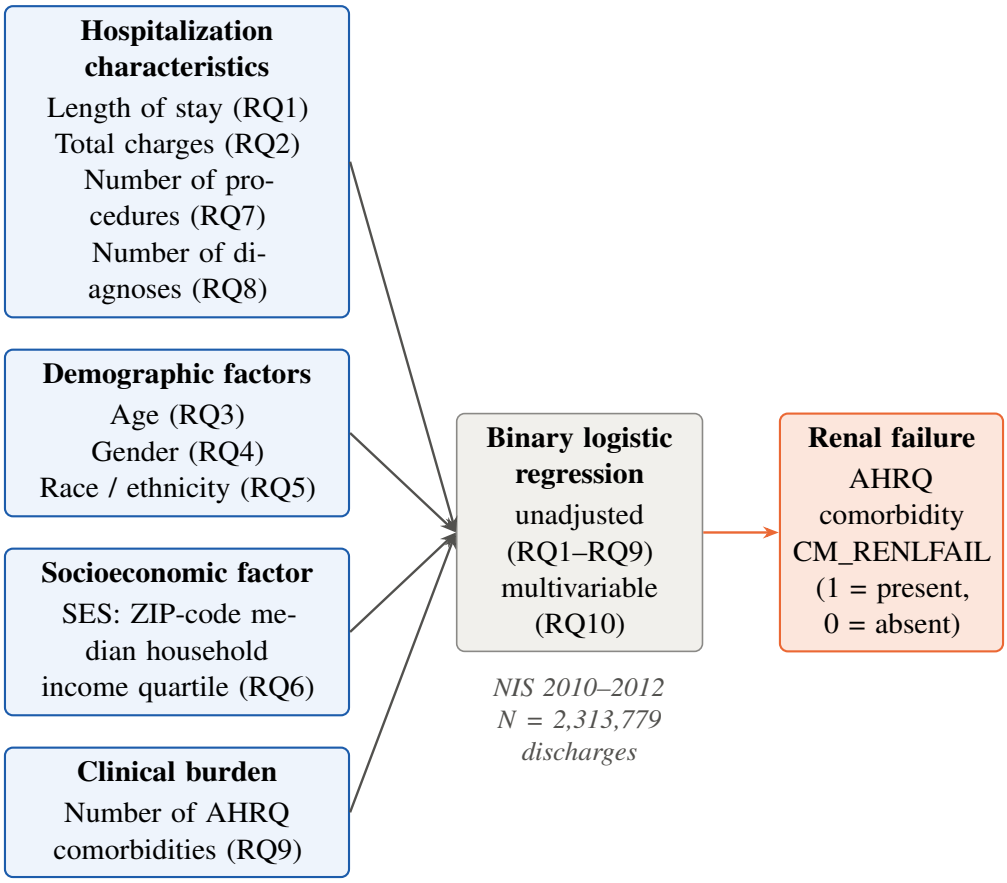


Figure 1

Conceptual Framework Linking the Nine Independent Variables and Ten Research Questions to Renal Failure

Note. Each independent variable is first tested individually against renal failure (RQ1–RQ9); all nine are then entered together into a single multivariable model (RQ10). Prepared from the research questions in Chapter I and the operationalization in Table 1.

literature. As a result, this study may close a knowledge gap, and its conclusions may be useful to academics and researchers in clinical research, critical care, and CKD studies.

This study might have practical uses as well. In order to slow the progression of renal failure, McQueen et al. (2017) emphasized the significance of early detection and clinical intervention for patients with diabetes who have CKD. Nephrologists may be better able to meet the needs of their patients if they have a better understanding of the differences between sexes in the etiology, mechanisms, and epidemiology of renal failure, according to Carrero et al. (2018). To address disparities in renal failure patients' access to care and outcomes, Carrero et al. recommended addressing societal factors as well as variations in the underlying pathophysiology of the disease. According to Kim et al. (2018), doctors should concentrate on restoring renal function and closely monitoring patients with risk factors for developing CKD. Therefore, CKD patients, nephrologists, physicians, and the local communities could all benefit from the study's findings.

Summary

The purpose of this study is to investigate the relationship between renal failure and length of stay, total charges, and their association with age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. The research questions and hypotheses addressed in this study (RQ1–RQ10) are stated in full in the Hypotheses section above and summarized in Figure 1.

Chapter 2 includes a comprehensive literature review with relevant research and theories related to the research questions. Chapter 3 will explain the research methodology, including data collection procedures, instrumentation, and data analysis processes used in the study. Chapter 4 will provide the results of the analysis, and Chapter 5 will detail how the results of this study support or do not support studies discussed in the literature review.

CHAPTER II: LITERATURE REVIEW

Chronic kidney disease (CKD). CKD is common and has emerged as one of the leading non-communicable causes of death globally. Statistics estimate that CKD affects an increasing number of people over time and further rises in importance among the various global causes of death (Zsom et al., 2022). CKD affects populations in different regions of the world unequally, likely due to differences in population demographic characteristics, comorbidities, and access to healthcare resources (Yu et al., 2019). The common nature and devastating effects of CKD should prompt major efforts to develop and implement effective preventative and therapeutic measures aimed at lowering the development of CKD and slowing its progression (Wilkinson et al., 2022). The problem to be addressed is renal failure and its association with age, gender, race, SES, number of procedures, insurance type, comorbidities, length of stay, admission type, risk of mortality, type of CKD, illness severity, number of chronic conditions, in a cohort of inpatients in the United States. The specific issue is that among inpatients with CKD, it is still unclear how renal failure relates to factors such as length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities. The purpose of this study is to investigate the relationship between renal failure and length of stay, total charges, and their association with age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. The review of literature section is organized into the following sections: chronic kidney disease (CKD), types of kidney diseases, economic consequences of chronic kidney disease, comorbidities in chronic kidney disease, the association between age and prevalence of chronic kidney disease, socioeconomic status and prevalence of chronic kidney disease, and an association between race and prevalence of chronic kidney disease.

The organization of the review is mapped in Figure 2.

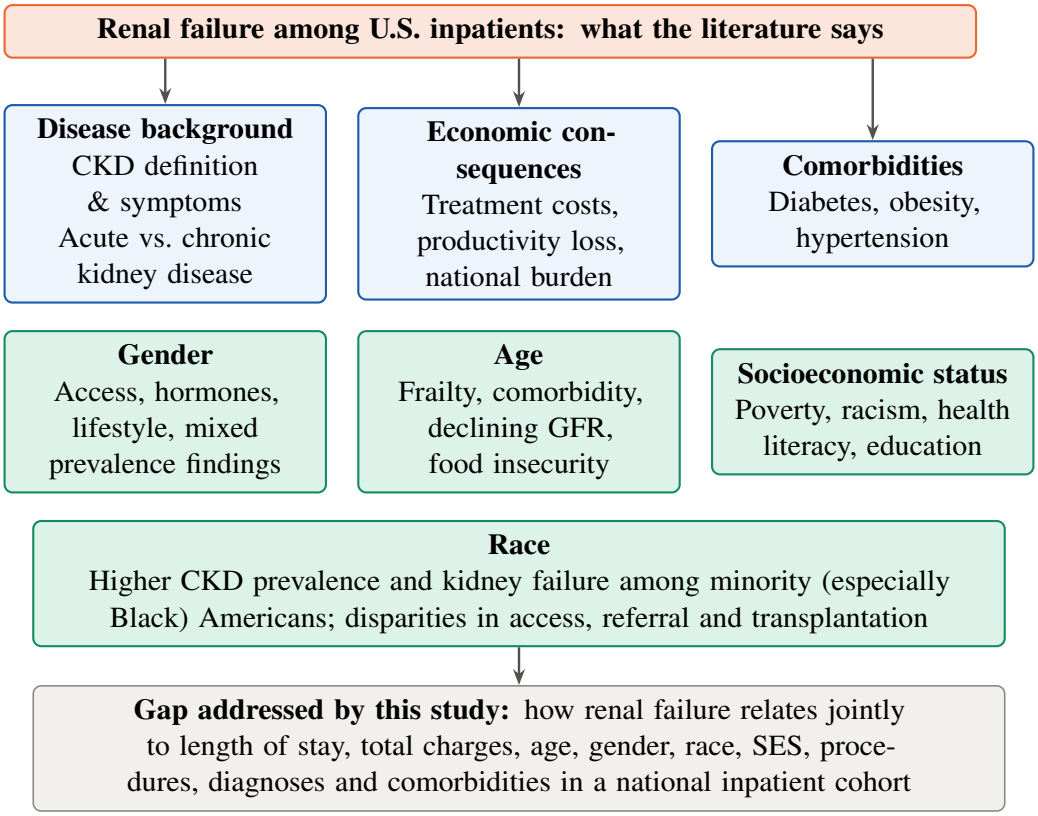


Figure 2

Thematic Structure of the Literature Review

Note. Blue boxes are contextual themes; green boxes correspond to predictors tested in this study. Prepared from the organization of Chapter II.

Review of Literature

Chronic Kidney Disease (CKD)

Chronic kidney disease (CKD), commonly known as chronic kidney failure or renal failure, involves a steady and gradual loss of all kidney functions (Saunders et al., 2020). This leads to impairment and difficulties in blood filtration or the removal of excess fluid from the blood (Brar & Markell, 2019). The accumulation of excess fluid in the blood leads to the failure of major key body functions. CKD has several symptoms and signs as a common and typically occurring clinical condition. Most important is to emphasize that the gradual loss of kidney functions results in fluid, body waste, or electrolyte problems (Hecking et al., 2022). Depending on its severity and progression stage, the following are some key symptoms of CKD, nausea, vomiting, loss of appetite, fatigue, and general body weakness (García et al., 2022). The other vital symptoms include sleep problems, increased rate of urinating, impairment in mental sharpness, consistent muscle cramps, and swelling of feet and ankles (Huang et al., 2019). In addition, dry and itchy skin, high blood pressure, shortness of breath, and chest pain are other common symptoms of renal failure (Saunders et al., 2020). Despite the above list of key characteristics of CKD, it is conventionally not agreed upon which of them would specifically rule out other conditions with similar features to CKD (Hu et al., 2020). The general complications and signs of CKD are not specific as they share with other diseases. Other illnesses may also exemplify similar illnesses or characteristics. It is, therefore, important to underscore that the symptoms of CKD are non-universal.

Types of Kidney Diseases

Conventionally, there are two types of kidney diseases currently known. The two include acute and chronic kidney injury (Evenepoel et al., 2021). Acute kidney injury occurs in the short term, while chronic kidney injury occurs longer. It is easier for people to recover from acute kidney injury with proper care management than from chronic kidney injury (Hounkpatin et al., 2020).

Acute kidney injury refers to the temporal or permanent loss of key functions, which could last less than three months. To address acute kidney injuries, patients may be recommended for dialysis (Hsu et al., 2021). Patients will likely recover from acute kidney injury and continue normally (Hu et al., 2020). In the instances where the damage to the kidney is significant, the condition might progress into lifelong kidney failure (Vart et al., 2020). Given the potential impact of acute kidney injury on one's future health, it is recommended that proper care is sought in the initial stages to prevent chronic kidney injury (Kovesdy, 2022).

The second type of CKD is chronic kidney disease. This condition occurs when kidneys are damaged for over three months, and the conditions are irreversible. Diagnosis of chronic kidney disease is usually based on the condition persisting beyond three months (Lai et al., 2019). While an individual can lead a typical life with chronic kidney diseases, they are likely to experience a gradual decline in their kidney function. This will eventually lead to the replacement of kidney therapy (Levey et al., 2020). The patients may be recommended for kidney transplants or dialysis. Researchers have identified several potential risk factors for chronic kidney diseases (Hu et al., 2020). High blood pressure also affects the kidney or leads to chronic kidney failure (Banerjee et al., 2019). It is therefore recommended that individuals seek timely screening of their bodies to identify the type of kidney failure they experience and organize for treatment to mitigate the unmonitored progression of the condition into chronic status (Hu et al., 2020).

CKD has emerged as one of the major causes of death in the 21st century (Hecking et al., 2022). Attributed partly to the rise in its risk factors, including obesity and diabetes, individuals suffering from CKD gradually increased to nearly 1.8 million as of 2019 (Huang et al., 2019). While the mortality of CKD has declined recently, the burden of the disease remains high and of great concern both among scholars and individuals (Huang et al., 2019; Hecking et al., 2022). Given its potential economic, social, and emotional impact on individuals, there have been concerns about investigating this condition in detail and accurately documenting its impact on people (Hecking et al., 2022).

Figure 3 contrasts the two types of kidney disease and their common end point, renal failure.

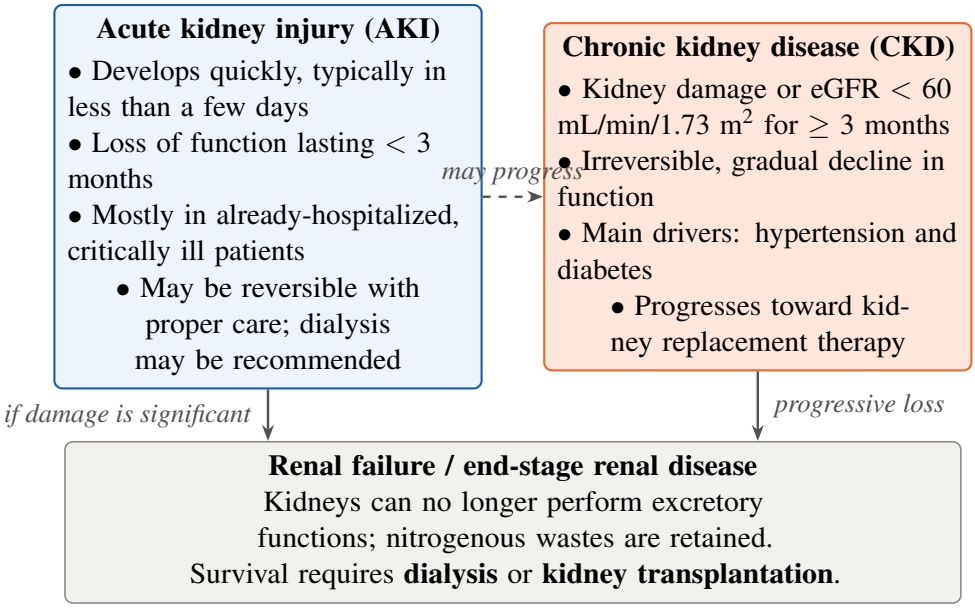


Figure 3

Acute Kidney Injury, Chronic Kidney Disease, and Progression to Renal Failure

Note. Synthesized from Chapter I and the Types of Kidney Diseases section (Evenepoel et al., 2021; Hu et al., 2020; Lai et al., 2019; Levey et al., 2020; National Kidney Foundation, 2016; Vaidya & Aeddula, 2022).

Economic Consequences of Chronic Kidney Disease

Chronic Kidney disease (CKD) refers to the gradual but continued loss of kidney functions over time. Bikbov et al. (2020) asserted that CKD develops as a result of the failure of the kidney to filter wastes and excess fluid from the blood. Failure by the kidney to filter wastes exposes the body to illnesses and medical conditions such as high blood pressure, weak bones, nerve damage, and poor nutritional health (Elshahat et al., 2020). The consequential impacts of CKD are the costs of treatment which have had severe economic impacts on developed and developing countries as well as individuals diagnosed with CKD or with family members diagnosed with CKD. Elshahat et al. (2020) systematically reviewed the literature on the impacts of CKD and found that at least 10% of the global population has been diagnosed with CKD, significantly stretching the health systems of many countries.

In the United States, 1 in every ten adults is diagnosed with CKD (Nguyen-Thi et al., 2021). A report published by the American Society of Nephrology revealed that nearly 26 million Americans had been diagnosed with chronic kidney diseases, increasing the risk of developing kidney failure (Thurlow et al., 2021). Another 20 million Americans were at elevated risk of developing kidney diseases (Kovesdy, 2022). Kovesdy (2022) further stated that over the last two decades, the number of American citizens aged 60 and 70 diagnosed with CKD and other kidney diseases has doubled and is expected to continue increasing as treatment costs increase. Shlipak et al. (2021) stated that the annual cost of treating end-stage renal disorders is more than $32 billion, with the costs being slightly over one-quarter of the allocated Medicare expenditures.

Examining the global economic burden of chronic kidney diseases, Nguyen-Thi et al. (2021) established that the annual treatment costs attributed to chronic kidney diseases are approximately $49 billion. In Italy, Nguyen-Thi et al. (2021) stated that 9.6% of patients were diagnosed with CKD, with the treatment costs totaling around 1.81 billion euros. Similar findings were reported in China and Thailand, where 77.5% of the patients on dialysis accumulated a total cost of $16,625 per year (Nguyen-Thi et al., 2021). Reiterating the findings of Nguyen et al. (2021), Elshahat et al. (2020) reported that the progression of CKD stages from 1 through 4 and 5 increased treatment costs by 1.3 ~ 4.2-fold. Elshahat et al. (2020) asserted that CKD progression to stage 4 increased the treatment costs for end-stage renal disease from $20, 110 to $ 100,593 for each patient. Overall, research has demonstrated that as CKD progresses, the costs of treatment increase while decreasing the economic well-being of people and the country as a whole.

Among CKD's key economic impacts is increased living costs for the general population. Walbaum et al. (2021) investigated CKD's health and economic impacts using data collected from six states in Chile and the United States. The quantitative analysis of the collected data revealed that an increase in the number of people diagnosed with CKD increased the costs of treatment, which significantly and negatively influenced their economic well-being through

increased living costs. With limited annual income and continued increase in treatment, people feel economically stretched and incapable of affording basic needs. Previously, Freeman et al. (2019) systematically reviewed 95 studies on CKD's economic and human impacts in the United States. The review results revealed that the increased severity of CKD reduced patients' quality of life by stretching their financial capabilities and increasing their living costs. In the preceding review, it is evident that the high costs of living, which is a characteristic of stretched and poorly functioning economy, is exacerbated by the high costs of CKD treatment.

Research has demonstrated that CKD diminishes the productivity of individual patients exposing their families to poverty and poor quality of life. Cloutier et al. (2020) investigated the societal burden of kidney disease in the United States. Cloutier et al. (2020) established that the United States experienced an annual loss of $ 1.4 billion in direct costs and $125 million in non-healthcare costs associated with CKD and autosomal dominant polycystic kidney disease. The indirect and non-healthcare costs are associated with loss in productivity due to unemployment, increased hospital, and homecare costs, and reduced productivity at the workplace. Similar results were reported by Brown and Elliott (2021), who researched CKD's basics and impacts on American citizens. Collecting and analyzing quantitative data from 425 CKD patients, Brown and Elliott (2021) reported that CKD disproportionately affected people with poor social determinants of health increasing poverty and low quality of life. Families with poor economic outcomes or living in poor environments experienced difficulties accessing high-quality, expensive care and thus suffered from high mortality and morbidity due to CKD.

Increased comorbidity, poverty, and low quality of life associated with CKD have significantly influenced and stagnated the economy of nations around the globe. Elshahat et al. (2020) illustrated that CKD's social and health effects interfered with their productivity and contribution to the economy, as much of the income was used for CKD treatment and associated comorbidities such as hypertension and diabetes. Research conducted by Cloutier et al. (2020) revealed that CKD cripples a nation's productivity as CKD patients cannot work effectively, causing loss of employment, premature mortality, and high healthcare costs that

stretch the economic limits of governments. Brown and Elliott (2021) demonstrated that CKD had worsened the social determinants of health needed by vulnerable and marginalized communities. Therefore, addressing such communities' social, economic, and health needs takes a significant economic boost by the government that interferes with its normal operations.

In earlier research, Lv and Zhang (2019), while examining the prevalence of and burden of CKD, established that the high costs of treatment and associated chronic diseases such as hypertension, aging, and diabetes mellitus increased the financial burden experienced by countries both poor and wealthy. Comparably, Morton and Shah (2020) stated that CKD had stunted the growth of many countries. Due to high healthcare costs, many middle-income and developing nations have diverted a chunk of their budgets to ease the burden of medication associated with CKD and other chronic illnesses experienced by their citizens, thus doing very little to grow their economies. Therefore, it is evident that CKD results in a substantial financial burden not only for developing nations but also for most wealthy nations around the globe. Therefore, there is a need for cost-effective preventive strategies to minimize the burden of kidney diseases.

The burden figures reported in the studies reviewed above are brought together in Figures 4 and 5.

Comorbidities in Chronic Kidney Disease

Research has demonstrated that CKD is associated with comorbidities that increase its severity and burden. Jacob et al. (2019) researched the current status of CKD, including causes and comorbidities, and established that hypertensive nephropathy and diabetic nephropathy were the leading causes of CKD. Further, retinopathy, diabetes, hypertension, cardiovascular disease, and chronic obstructive pulmonary disease were common CKD-associated comorbidities. Sullivan et al. (2020) investigated the association between multimorbidity and severe clinical outcomes in patients with CKD. The systematic review of 21 studies revealed that CKD development was increased in patients at risk of multimorbidity, including cardiovascular diseases that accelerated the progression of kidney failures. Like Sullivan et al. (2020), MacRae et al. (2020) stated

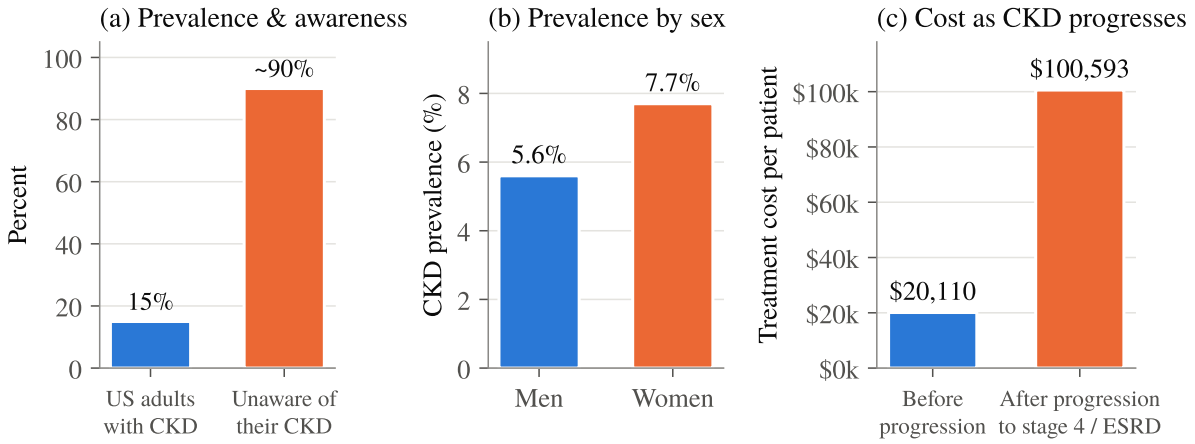


Figure 4

Selected Prevalence and Cost Statistics on Chronic Kidney Disease Reported in the Literature

Note. Panel (a): Centers for Disease Control and Prevention (2022); panel (b): Neugarten and Golestaneh (2019), U.S. survey 2005–2010; panel (c): Elshahat et al. (2020). Values are as reported by the cited authors; they are not estimates from the present study sample.

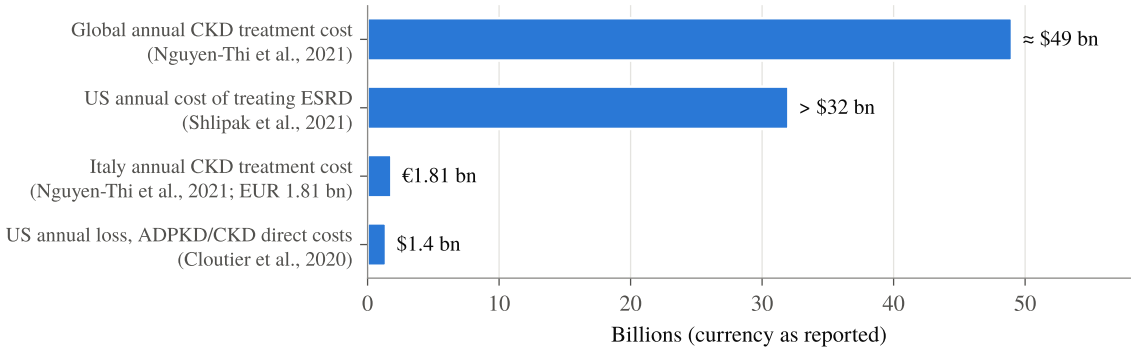


Figure 5

Annual Economic Burden of Chronic Kidney Disease Reported in the Literature

Note. Bars show values in billions of the currency reported by each source (U.S. dollars, except Italy in euros). ADPKD = autosomal dominant polycystic kidney disease; ESRD = end-stage renal disease.

that CKD is associated with cardiovascular diseases, hypertension, and diabetes. MacRae et al. (2020) explained that being the leading cause of mortality and morbidity globally, CKD is common to people with underlying medical conditions such as diabetes, hypertension, and cardiovascular disease due to deteriorated kidney functions.

Diabetes and Chronic Kidney Disease

People with diabetes are at an increased risk of developing chronic kidney disease. Centers for Disease Control in 2019 published that in every three adults with diabetes, 1 has CKD. Per the CDC (2019) reports, people diagnosed with type 1 and type 2 diabetes are at risk of developing kidney infections as both forms of diabetes have been reported as risk factors for kidney diseases. Extending the conclusions drawn by the CDC (2019), Kovesdy (2022) investigated the relationship between diabetes and chronic kidney disease and established that diabetes accounted for 44% of new cases of kidney diseases in the United States. Diabetes damages the blood vessels throughout the body, interfering with the kidneys, nerves, heart, muscles, and skin. In concert with Kovesdy (2022), de Boer et al. (2022) stated that high blood sugars damaged the kidney's nephrons, interfering with their filtering work and thus accumulating toxic wastes that gradually damage the kidneys. Therefore, as a comorbidity, damage to small blood vessels supply the kidney, and the nephrons contribute to CKD.

Supporting the findings that CKD develops due to the damage caused to small blood vessels and nephrons of the kidney by diabetes, Jitraknatee et al. (2020) investigated the prevalence and risk factors for CKD among patients with type 2 diabetes. In the cross-sectional research with 1096 primary care patients, Jitraknatee et al. (2020) conducted a multivariate logistic analysis of the data and established that type 2 diabetic patient were at an increased risk of developing CKD, given the complications associated with glycemic control. Corsonello et al. (2020) reported consistent results examining the multimorbidity pattern of chronic kidney disease. The data collected from 2252 patients indicated that older adults diagnosed with diabetes were at elevated risks of developing CKD, congestive heart failure, and hypertension. Corsonello et al. (2020) further established that the multimorbidity pattern observed in patients

resulted from the severity of CKD that exposed patients to respiratory and neurologic disease and hypertension.

Diabetes, both type 1 and type 2, has been associated with immune system dysfunction, increasing the risks of CKD and cardiovascular diseases. Berbudi et al. (2020) investigated the impacts of type 2 diabetes on the immune system and explained the hyperglycemia dysfunctions of the immune system, making it difficult for the body to control the spread of disease-causing pathogens. That is, type 2 diabetes increases the susceptibility of individuals to infections. Similar results were reported by Daryabor et al. (2020), who studied the impacts of type 2 diabetes on the immune system and body organs. Dysfunction of the immune system increases the body's susceptibility to infection, and the damage to blood vessels results in organ failure. Associated with kidney disease, Daryabor et al. (2020) demonstrated that increased oxidative stress and organ failure contributed to the development and progression of chronic kidney disease. Thus, both researchers concur that the development of CKD among diabetic patients is associated with a dysfunctional immune system and organ failure.

Several scholars supported findings similar to those reported by Daryabor et al. (2020). Pilemann-Lyberg et al. (2019) investigated the decline in kidney functions because of uric acid and cardiovascular disease in patients with type 1 diabetes. Quantitatively analyzing data from 670 patients with type 1 diabetes, it was established that higher Uric acid levels were associated with a significant decline in kidney functions and subsequent development of CKD. Bramlage et al. (2019), while analyzing the outcome of the data collected from a large dataset with type 2 diabetes patients, chronic kidney disease was common among studied patients, followed by hypertension. Comparably, conducting a large multinational cohort study with a total of 1,177,896 patients with type 2 diabetes, Birkeland et al. (2020) established that chronic kidney disease was prevalent among such patients and increased their risks of developing cardiovascular disease and mortality. Overall, researchers have evidence that diabetes increases the chances of CKD.

Diabetes is another risk factor for CKD. Studies examining CKD prevalence have linked CKD to diabetes. For instance, research conducted in the United States in 2019 revealed the prevalence of CKD in stages three to four as 14%. The relationship between diabetes and CKD prevalence was also reported in another meta-analysis that identified diabetes as a major contributing factor to CKD. This study provides enough empirical evidence linking diabetes to CKD.

Obesity and Chronic Kidney Disease

Obesity has increased the risks of developing kidney disease, including chronic kidney disease. Kovesdy et al. (2019) stated that obesity is directly associated with the development of end-stage renal disease and chronic kidney disease. Kovesdy (2019) asserted that being obese or overweight increases the risks of high blood pressure and diabetes, which are the key factors of kidney diseases. Friedman et al. (2021) further noted that in obese people, the intraglomerular pressure and compensatory hyperfiltration increase to meet the metabolic demands damaging the kidneys and increasing the risks of developing CKD. Nawaz et al. (2022) contended that increased intraglomerular pressure led to nephrolithiasis and kidney cancer that contribute to the progression of kidney disease to CKD. Therefore, obesity increases the risk of developing CKD.

Obesity exerts significant pressure on the kidneys, which have to meet the metabolic demands of obese people, especially regarding the filtration of wastes. Kovesdy et al. (2019) revealed that the high metabolic rates associated with obesity stretched the functioning of the kidney or damaged them together, leading to CKD. Extending the research done by Kovesdy et al. (2019), Than et al. (2020) examined the role of obesity in developing chronic kidney disease and cardiovascular complications. The findings indicated that obesity increased the risks of CKD and obesity-related glomerulopathy (ORG) that negatively influenced the functioning of the kidneys. Than et al. (2020) established that in obese patients, insulin resistance, changes in the renal hemodynamics, and the activation of sympathetic and renin-angiotensin-aldosterone systems increased the risks of developing CKD. These changes impair the kidney's normal functioning, increasing the risk of developing CKD.

The body weight's pressure on the kidneys impairs their functions and exposes obese persons to kidney diseases. Garcia-Carro et al. (2021) explored what nephrologists thought about obesity and the development of chronic kidney diseases. Similar to Than et al. (2020), Garcia-Carro et al. (2021) established that obesity was associated with obesity-related glomerulopathy associated with renal hemodynamic changes leading to albuminuria and hyperfiltration that impairs the glomerular filtration exposing individuals to CKD. In concert with Kovesdy et al. (2019), Garcia-Carro et al. (2021) established that obesity was associated with diabetes and hypertension, many people's leading causes of CKD. Therefore, in addition to ORG and renal hemodynamics attributed to CKD, obesity increases the risks of hypertension and diabetes, leading causes of chronic kidney disease.

Hypertension and Chronic Kidney Disease

Hypertension has been demonstrated as the leading cause and outcome of CKD. While investigating the management of hypertension in chronic kidney disease, Pugh et al. (2019) illustrated that increased stiffness of the artery, upregulation of the renin-angiotensin-aldosterone system that promotes retention of salt and water as well as increased sympathetic tone have been associated with the development of hypertension in patients with CKD. Weldegiorgis and Woodward (2020) investigated the impacts of hypertension on chronic kidney disease and end-stage renal disease in men and women. Systematically reviewing six cohorts of the retrieved studies, the researchers demonstrated that hypertension increased the risks of CKD in men and women at rates higher than other risk factors such as diabetes and cardiovascular disease. In concert with Pugh et al. (2019), Weldegiorgis and Woodward (2020) contended that hypertension increased sympathetic tone and changed renal hemodynamics, increasing the risks of developing CKD and ESRD.

As the leading cause of CKD, hypertension increases the blood pressure of the kidney vasculature, increasing the risks of damage and development of CKD. Ku et al. (2019) stated that uncontrolled high blood pressure in the kidney increases intraglomerular pressure and impairs the glomerular filtrate's functioning and glomerular filtration. Damage to the glomerular

walls increases the filtration of proteins in urine, the first sign of CKD. High blood pressure due to hypertension damages the kidney's blood vessels, impairing its functioning and ability to filter wastes from the blood, leading to CKD (Pugh et al., 2019). Georgianos and Agarwal (2020) also investigated the prevalence and treatment of resistant hypertension in CKD and found that resistant hypertension was prevalent in patients diagnosed with CKD. Townsend (2020), while reviewing the literature on the pathophysiology of hypertension in CKD, established that high blood pressure in the kidney impaired its ability to maintain a balance between electrolytes, endothelial functions, and hormones.

Figure 6 synthesizes the comorbidity pathways reviewed in this section.

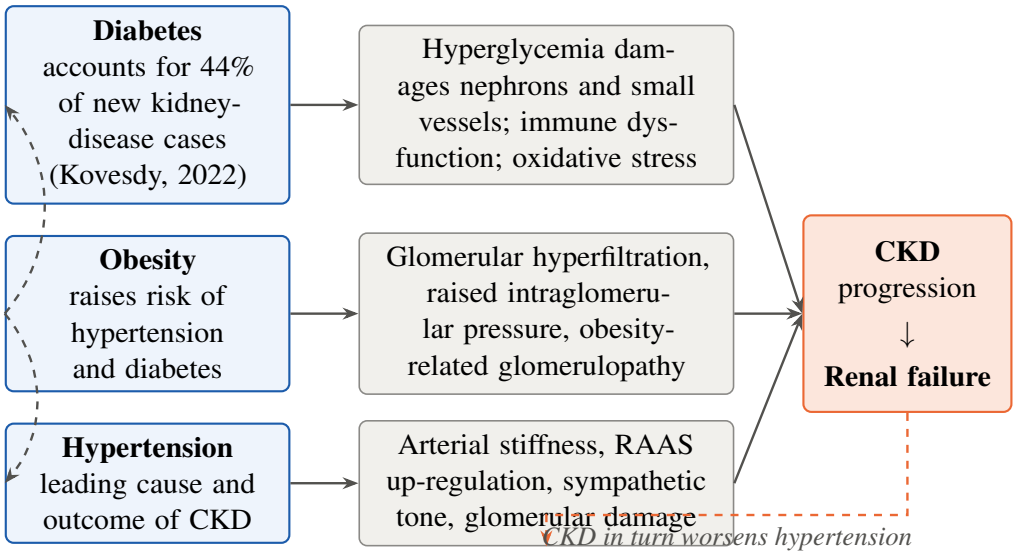


Figure 6

Comorbidity Pathways to Chronic Kidney Disease and Renal Failure Described in the Literature

Note. Dashed arrows show that obesity increases the risk of both diabetes and hypertension. Synthesized from de Boer et al. (2022), Daryabor et al. (2020), Friedman et al. (2021), García-Carro et al. (2021), Kovesdy (2022), Kovesdy et al. (2019), Ku et al. (2019), Pugh et al. (2019), and Than et al. (2020).

Association between Gender and Prevalence of Chronic Kidney Disease

There are gender disparities in access to kidney health that have increased cases of chronic kidney disease. Tong et al. (2022) studied the perspectives on gender disparities in access to kidney healthcare and reported that chronic kidney disease was more prevalent in women than

men as there was limited access to kidney healthcare by women due to stereotyping by clinicians and economic disadvantage. Garcia et al. (2022) also examined the gender differences in access to kidney healthcare globally and established that women received fewer kidney healthcare services than men because of a lack of medical insurance coverage, gender stereotyping in health centers, and cultural and social norms. In a similar study, Vinson (2022) investigated the effects of gender disparities in access to kidney transplants and reported that women with chronic kidney disease were less likely to be referred to a transplant program compared to men as a consequence of gender disparities in kidney health service provision hence the high prevalence among women. The prevalence of chronic kidney disease cases has increased due to gender differences in access to kidney health.

Gender disparities in access to kidney healthcare affect women more than men hence the prevalence of chronic kidney disease among women. Katz-Greenberg and Shah (2022) examined the gender disparities in kidney transplants by sex and found that women were less likely to be referred to a donor or kidney transplant program than men resulting from gender stereotyping and limited economic power by women to access kidney treatment. Salas et al. (2022) also studied the gender disparities in kidney transplants and reported that men were more likely to be referred to a donor or a transplant program, and receive a kidney transplant than women, hence the prevalence of chronic kidney disease in women. Consistent with Katz-Greenberg and Shah (2022) and Salas et al. (2022), Smothers et al. (2022) investigated the impacts of gender on access to kidney transplants, and dialysis referral between men and women and established women were less likely to be referred to a donor or a transplant program as compared to men thus the increased prevalence of chronic kidney diseases among women. Women are more likely than men to suffer from gender inequities in access to kidney healthcare, contributing to the higher prevalence of chronic kidney disease in women.

Women's lack of access to comprehensive healthcare has contributed to the increased cases of chronic kidney disease. Patzer et al. (2020) studied the variations in dialysis referrals for patients suffering from chronic kidney disease and reported that women and older persons

were less likely to be referred for dialysis hence the high prevalence of chronic kidney disease. Previously, Kim et al. (2019) examined the trends in kidney transplant referrals for patients with chronic kidney disease and found that women were less likely to be added to the waitlist for donors due to gender stereotyping in kidney health provision. In addition, Melk et al. (2019) investigated how sex and gender affected organ transplantation and reported that women with chronic kidney disease were less likely to be considered for kidney transplantation than men due to sex discrimination towards women and the prevalence of psychosocial concerns regarding transplantation. The rise in the incidence of chronic renal disease has been attributed to women's lack of access to comprehensive healthcare.

Estrogen in women and testosterone in men is vital in spreading and preventing chronic kidney disease. Singh et al. (2022) investigated the importance of estrogen in preventing chronic kidney disease and established estrogen to be protective against chronic kidney disease in women. In a similar study, Van der Burgh et al. (2022) studied the role of testosterone in kidney function and the clinical outcome of chronic kidney disease and found that low testosterone levels were linked to a higher prevalence of chronic kidney disease in men compared to women. In a previous study, Zhao and Schooling (2020) examined the role of testosterone in chronic kidney disease in men and women and reported that bioavailable testosterone in men was linked to low glomerular filtration rate and higher albuminuria which are associated with high chronic kidney disease prevalence. Hormones play an important role in the management and prevalence of chronic kidney disease.

Gender hormones significantly influence chronic kidney disease management and prevalence. Li et al. (2021) examined the impacts of the high circulating follicle-stimulating hormone level as a potential risk factor for renal dysfunction and reported that a decrease in estrogen levels in women during menopause was associated with an increase in renal dysfunction and a high prevalence of chronic kidney disease. Sharma et al. (2020) also investigated the influence of testosterone replacement therapy in men and found that testosterone deficiency was a risk factor for chronic kidney disease prevalence and was linked to suppressed kidney functioning.

Like Sharma et al. (2020), Brar and Markell (2019) studied the impact of hormonal differences and gender disparities in patients with renal diseases and reported estrogen in women had an anti-fibrotic influence and thus suppressed chronic kidney disease in women compared to testosterone which had a pro-inflammatory effect thus increased chronic kidney disease effects in men. Different hormones and hormonal changes in men and women affect chronic kidney disease's prevalence.

Different lifestyles between women and men have contributed differently to the prevalence of chronic kidney disease. Al-Rousan et al. (2022) investigated the trends in excessive alcohol consumption in men and women and reported that men were more into excessive alcohol consumption, which was associated with negative health outcomes such as chronic kidney disease and liver cirrhosis. Previously, Huang et al. (2019) examined the association between gender and lifestyle in the prevalence of chronic kidney disease and found that chronic kidney disease cases were high in men who were serial smokers and binge alcohol consumers. Consistent with Al-Rousan et al. (2022) and Huang et al. (2019), Hecking et al. (2022) studied the risk factors in the prevalence of chronic kidney disease in men and reported unhealthy lifestyles by men such as drug use, alcohol consumption, and unhealthy dietary patterns have increased the frequency of chronic kidney cases. The prevalence of chronic kidney disease has been influenced differentially by men's and women's lifestyle choices, such as binge drinking in men and unhealthy diets in women.

Gender lifestyle differences influence the trends in chronic kidney disease between men and women. Hu et al. (2020) examined the incidences of chronic kidney disease in trends of alcohol consumption among men and women and reported higher cases of chronic kidney diseases characterized by excessive drinking by men compared to women with a moderate drinking lifestyle. In an earlier study, Lai et al. (2019) studied the link between alcohol consumption and the risk of chronic kidney disease and established that high alcohol consumption was associated with the onset of chronic kidney disease among men and women. In a different study, Banerjee et al. (2019) investigated the link between adherence to a dietary program and the prevalence of

chronic kidney disease and established that women were more likely to adhere to a diet program than men hence a higher prevalence of chronic kidney disease in men. Trends in chronic kidney disease affect both men and women differently depending on gender and lifestyle choices.

Researchers have demonstrated differences between gender and the prevalence of chronic kidney diseases. Kristensen et al. (2019) conducted a qualitative study to examine how gender influences the prevalence of CKD and found that women who develop pre-eclampsia condition during pregnancy are at risk of developing kidney dysfunctions. Agreeing with Kristensen et al. (2019), Neugarten and Golestaneh (2019) also surveyed between 2005–2010 to investigate the relationship between gender and the prevalence of CKD and reported that more women were prone to CKD than men (7.7% versus 5.6%). Kovesdy (2022) also examined how gender affected the prevalence of CKD and established that women are more likely to suffer CKD than men because of vulnerability to recurrent urinary tract infections. Studies reviewed indicate that CKD is more prevalent in women than men.

Gender impacts the prevalence of CKD. Duan et al. (2019) conducted a cross-sectional survey with 23869 participants to examine how gender affected CKD incidence rates and reported that men are at a higher risk of kidney failure because they lack the estrogen hormone that protects the kidney like their women counterparts. Cañadas-Garre et al. (2019) also conducted a qualitative study to explore the spread of CKD among men and established that men are at risk of renal dysfunction more than women because of high testosterone hormones. Concurring with Duan et al. (2019) and Cañadas-Garre et al. (2019), Valdivielso et al. (2019) investigated how sex influenced the prevalence of CKD and reported that males with enlarged prostate that causes a blockage to their urinary tract might experience loss in kidney function. Therefore, consistent evidence across the three studies shows that men are more prone to CKD than women.

Gender differences determine the rate of CKD occurrence through a systematic literature review. Valdivielso et al. (2019) investigated CKD infection rates among men and women and found that women are more prone to CKD than men because of endometrial cysts that may

block the renal capsule surrounding the kidneys leading to renal failure. Deschamps-Lenhardt et al. (2019) reported contradictory findings; they conducted a meta-analysis to examine the role of hormones in CKD among women and found that women are less likely to suffer CKD until menopause because of high levels of estrogen that hinder the progression of renal diseases. Al Khalaf et al. (2022) conducted a systematic literature review to analyze how gender differences impact CKD rates in a population and found that women are more exposed to CKD than men because of postnatal complications such as high blood pressure. Articles reviewed thus far reveal that women are more susceptible to CKD than men.

Men and women have different susceptibility rates to CKD. In a qualitative study, Borghi et al. (2020) investigated the link between gender and vulnerability to CKD and highlighted that more women record an increased number of CKD cases than men because females lose a lot of blood and fluids during delivery, leaving them exposed to kidney failure. Wu et al. (2020) also conducted a meta-analysis study to examine the spread of CKD among adults and found that men have a higher protein intake than women, leading to increased cases of kidney failure. Evenepoel et al. (2021) also investigated the association between gender and the prevalence of CKD and highlighted that women have low levels of creatinine that males, which explains increased renal failure among females. Taking together the evidence reviewed thus, it can be concluded that sex differences influence CKD prevalence rates in a population.

Existing research has shown that the prevalence of CKD is higher in women than men. A study in the United States estimated that one in five females had CKD. The potential reasons for the differences in CKD prevalence across gender have been attributed to hormone differences (Duan et al., 2019). However, there are no conventionally agreed-upon reasons for the differences in CKD prevalence across genders (Neugarten & Golestaneh, 2019). Researchers have advocated for further research in this area to understand better the factors contributing to differences in CKD prevalence across genders. Despite the findings, some researchers have reported increased CKD across genders. For instance, Saunders et al. (2020) conducted a systematic review to investigate the prevalence of CKD among the elderly population in the

United States. The research findings indicated that females had higher chances of CKD than men, although there were no explanations for the differences (Cañadas-Garre et al., 2019; Valdivielso et al., 2019). Similar results were reported by Kristensen et al. (2019) in their study on the risk factors and prevalence of CKD among patients in nursing homes. Upon conducting the analysis, the investigators reported that CKD was prevalent among females compared to males. However, like previous studies, the researchers did not explain the cause of the difference in prevalence, needing further research.

There are also inconsistent results about the prevalence of CKD across genders. While some studies have reported a high prevalence of CKD in females, researchers such as Hecking et al. (2022) have reported contradicting findings. Specifically, Al Khalaf et al. (2022) found that while previous researchers had established a high prevalence rate of CKD among females, their results indicated that men were more vulnerable to CKD. Similar results were also reported by Deschamps-Lenhardt et al. (2019), whose findings also revealed that out of 215 patients who took part in the study, 60% were men with CKD symptoms. Although the population size was limited, the researchers concluded that CKD was more prevalent in males than females and recommended additional research to expand and replicate their findings. Other studies have established no significant difference in CKD prevalence across gender (Borghi et al., 2020). Researchers such as Wu et al. (2020) have reported that CKD affects all genders equally, and the difference reported by previous researchers could be attributed to population skewness. Previous researchers, such as Evenepoel et al. (2021), used non-transferable or nongeneralizable findings in other settings. Given the existing inconsistencies in research, there is a need to further this area and investigate the extent to which gender influences the prevalence of CKD.

The Association between Age and Prevalence of Chronic Kidney Disease

Frailty in old age is characterized by reduced muscle performance and organ failure and is prone to illness. Using secondary data, Kennard et al. (2022) investigated the impacts of frailty in old age persons and found that older adults suffering from flaws were more likely to have chronic kidney disease due to being vulnerable to health stressors such as multi-organ defects. In a

similar study, Chang et al. (2022) examined the impacts of frailty on the prevalence of chronic kidney disease in old-aged individuals and found that patients with chronic kidney disease suffered from frailty, evidenced by old age, comorbidity, and limited social care. Soldati et al. (2022) also studied the impact of frailty on patients undergoing hemodialysis and established higher frailty index in older people was typified by increased hospitalization due to chronic kidney disease. Frailty affects the nervous system and interferes with the functioning of vital organs like the kidneys, thus increasing susceptibility to chronic diseases in old age.

Older people with frailty are likely to have chronic and end-stage kidney disease. Hurst et al. (2022) examined the outcomes of older people living with defects and chronic kidney disease and found that deficiency in older populations was associated with frequent hospitalization of older people with chronic kidney disease and other non-communicable diseases. In a different study, Wilkinson et al. (2022) investigated the relationship between frailty and chronic kidney disease in elderly individuals. They reported that weakness was prevalent in older people with chronic kidney disease and had adverse outcomes such as propagating end-stage kidney disease and mortality. In addition, Otobe et al. (2022) studied the effects of frailty and physical performance of patients suffering from chronic kidney disease and found that deficiency in older people was likely to increase the prevalence of chronic kidney disease cases as a result of reduced organ function. Frailty interferes with body organs' ability to function, such as the kidneys, and hurts the neurological system, negatively impacting the health of older people and increasing the prevalence of chronic kidney disease in elderly populations.

Increased cases of chronic kidney diseases and mortality characterize frailty in older populations. Walker et al. (2023) studied the association of frailty and chronic disease in older individuals and established frailty increased incidences of chronic kidney diseases among old-aged populations. Walker et al. (2023) also reported that frailty in older people was associated with low glomerular filtration rates and high albuminuria, leading to the high prevalence of chronic kidney disease in older individuals. In a previous study, Mei et al. (2021) examined frailty as a sign of adverse outcomes in older patients suffering from chronic kidney disease

and reported that frailty was a predictor of negative effects of chronic kidney disease, such as mortality and increased hospitalization. Like Walker et al. (2023), Nixon et al. (2020) investigated the link between frailty and chronic kidney disease in older populations and reported that frail patients were susceptible to adverse symptoms of chronic kidney disease and worse health-related quality life. Increased mortality and chronic renal disease incidences are frailty characteristics in elderly populations.

Comorbidity has increased the prevalence of chronic kidney disease in older individuals. According to the Centers for Disease Control and Prevention ([CDC], 2019), diabetes and high blood pressure in older individuals were the main causes of chronic kidney disease; hence the prevalence of chronic kidney disease is increased by the presence of diabetes and high blood pressure in older people. Kovesdy et al. (2022) also studied the epidemiology of chronic kidney disease and found that chronic kidney disease was more prevalent in older patients suffering from diabetes mellitus and hypertension. In contrast, Miao et al. (2022) examined the symptoms of chronic kidney disease and mortality rates in older people and reported that older people in the United States had worse chronic kidney disease outcomes than China due to high comorbidity. In older people, chronic renal disease is more common due to comorbidity.

Underlying medical conditions in older people are one of the risk factors for the prevalence of chronic kidney disease. Argano et al. (2022) examined the impacts of diabetes mellitus and its comorbidities on the health outcomes of older patients and reported that diabetes was associated with various illnesses, including chronic kidney diseases, thereby increasing the prevalence of chronic kidney disease in older populations. Jiang et al. (2022) also investigated the adverse comorbid conditions among cancer survivors and reported that old age cancer survivors were at a high risk of developing chronic illnesses such as chronic kidney disease due to their weakened immune systems. In an earlier study, Novick et al. (2020) studied the effects of Covid 19 on older patients suffering from chronic kidney disease and established that the health effects of Covid 19 escalated chronic kidney disease outcomes and prevalence.

Comorbidity in older populations is highly characterized by the prevalence of chronic kidney disease among the elderly.

One of the risk factors for chronic kidney disease in elderly individuals is underlying medical conditions. Law et al. (2022) studied how hypertension was associated with chronic kidney disease and established that hypertension in older people was a major cause of chronic kidney disease, with increased left ventricle in the aging population being an indication of chronic kidney disease. Other researchers, such as Oliveros et al. (2020) and Ku et al. (2019), also studied the effects of comorbidity on chronic key disease. Oliveros et al. (2020) assessed the management and challenges of hypertension in older adults and reported that hypertension was widely prevalent in older individuals and was linked to various negative health outcomes such as chronic renal disease and renal failure. Ku et al. (2019) studied the links between hypertension and chronic kidney disease in older individuals and reported that hypertension was common in older patients; thus, sustained hypertension increased the prevalence of chronic kidney disease. The frequency of chronic renal disease among the elderly is a key indicator of comorbidities in older populations.

The glomerular filtration rate decreases with age, therefore, suppressing kidney functioning. Schrauben et al. (2020) examined the risk factors in the hospitalization of older patients with chronic kidney disease and reported low glomerular filtration rate was common amongst them, which affected the normal functioning of kidneys hence the increased prevalence of chronic kidney disease in older people. In addition, Cobb et al. (2020) investigated the factors associated with chronic kidney diseases in older individuals and established that older age was associated with a low glomerular filtration rate, impeding kidneys from functioning fully, thus increasing cases of chronic kidney disease in aged groups. In a different study, Delanaye et al. (2019) studied the diagnosis of chronic kidney disease in older adults and found that a low glomerular filtrate rate was linked to old age, thus increasing the prevalence of chronic kidney disease in older people. Chronic kidney disease among the older population is characterized by a reduced glomerular filtration rate which reduces the normal functioning of the kidneys.

A low glomerular filtration rate decreases the efficiency of the kidneys' functions. Schaeffner et al. (2022) studied glomerular filtration rates in older persons and their effects on chronic kidney disease and reported that glomerular filtration rates decreased with age, which hurt kidney health and increased chronic kidney disease. In a similar study, Zsom et al. (2022) examined the estimation of glomerular filtration rates on the outcome of chronic kidney disease in different age groups and found that older people had a decreased glomerular filtration rate than young individuals hence the prevalence of chronic kidney cases in older populations. However, Kanbay et al. (2019) examined the impacts of hyper glomerular filtration rates in older people and reported that increased glomerular filtration rates indicated chronic kidney diseases among the elderly. Lower glomerular filtration rates in older populations result in less effective kidney function hence susceptibility to chronic kidney disease.

Low glomerular filtration rates in older people have increased the prevalence of chronic kidney disease due to reduced kidney functioning. Eriksen et al. (2020) investigated the differences in glomerular filtration rates in middle and old-age populations and reported that old-age people had a lower glomerular filtration rate than middle-aged people hence the high prevalence of chronic kidney disease in older age groups. Previously, Lees et al. (2019) studied the effects of glomerular filtration rates as a predictor for cardiovascular and kidney disease and found that low glomerular filtration rates were prevalent in older populations and were associated with chronic kidney disease. Consistent with Eriksen et al. (2020) and Lees et al. (2019), Hirst et al. (2022) studied the effects of changes in glomerular filtration rates over time and age and reported an increase in the cases of chronic kidney diseases among the elderly highly characterized a decrease in glomerular filtration rates in older individuals. Limited glomerular filtration rates in older people are characteristic of chronic kidney disease.

Old age is highly characterized by a low intake of nutritious foods that promotes illnesses' weakening of the nervous system. Vargas-Vázquez et al. (2023) studied the effects of food insecurity in old patients suffering from chronic kidney disease and reported that older patients have multiple food insecurities that impede them from adhering to treatment routines for their,

thus increasing the prevalence of chronic kidney disease. Kalantar-Zadeh and Moore (2020) also examined the effects of food insecurity on patients suffering from chronic kidney disease during the Covid 19 pandemic and reported that food insecurity among the older population increased the prevalence of chronic kidney disease because of reduced nutrient intake. In a subsequent study, Te Vazquez et al. (2021) investigated the link between food insecurity and cardiometabolic conditions and established that food insecurity in adults was associated with obesity and hypertension, which increased the prevalence of chronic kidney disease. Low nutrient intake in old age contributes to negative health outcomes that impair the neurological system.

Food insecurity in old-age populations is a risk factor for chronic kidney disease spreading among the elderly. Leung et al. (2020) studied the association between food insecurity and health outcomes among older individuals and reported that food insecurity was associated with negative health outcomes such as chronic kidney disease, diabetes, and hypertension among the elderly. Like Leung et al. (2020), Leung and Wolfson (2021) examined the impacts of food insecurity on health outcomes in older adults and found that old age was characterized by food insecurity which negatively impacted health outcomes such as chronic bronchitis and renal diseases. In a different study, Kihlström et al. (2019) investigated the influence of food insecurity on quality health life among older adults and found that food insecurity was linked to negative health outcomes such as chronic kidney disease and hypertension among elderly individuals. Food insecurity in old-aged populations is highly associated with the prevalence of negative health outcomes such as chronic kidney disease.

Socioeconomic Status and Prevalence of Chronic Kidney Disease

Poverty. Poverty is one of the socioeconomic factors that affect the prevalence of chronic kidney disease. Nguyen et al. (2021) examined the prevalence of chronic kidney disease in high and low-poverty counties in the United States by using secondary data and found high poverty counties were highly characterized by increased chronic kidney diseases diagnosis because of less access to comprehensive healthcare and medical insurance compared to low poverty

areas. In a similar study, Boynton et al. (2022) studied the impacts of poverty on chronic kidney disease in children by using secondary data and established that low-income neighborhoods were typified by exacerbated rates of chronic kidney disease in children compared to high-income areas. In an earlier study, Norris and Beech (2021) investigated the influence of poverty on chronic kidney disease infections and reported that poverty was strongly associated with chronic kidney disease outcomes. Poverty is a major socioeconomic factor that influences the prevalence of chronic kidney disease by lack of medical insurance and access to healthcare.

Lack of access to social needs due to poverty has worsened chronic kidney disease. Morton (2022) studied the effects of socioeconomic factors on the increased rates of chronic kidney diseases worldwide and reported that countries with low income and high poverty levels had a higher prevalence of chronic kidney disease than countries with low poverty levels, as evidenced by inadequate screening and diagnosis of patients. Similar to Morton (2022), Novick et al. (2022) investigated the influence of unstable housing on the progression of chronic kidney disease and found that persons facing homelessness were more likely to contract chronic kidney disease than people with stable housing as they lacked medical insurance cover and access to kidney health services. In a previous study, Purnell et al. (2019) examined the impacts of poverty on donor kidney transplants and established that women living in impoverished areas were at a higher risk of graft loss and chronic kidney diseases than men in the same area. Poverty is a major socioeconomic factor that has escalated chronic kidney disease infections to limited healthcare access, such as albuminuria screening.

Poverty levels have a direct influence on the prevalence of chronic kidney disease. By conducting a qualitative study, Iorember and Bamgbola (2022) examined the influence of poverty on accessing kidney healthcare services by children and established poverty has led to a lack of medical insurance which has escalated the cases of kidney diseases among children from poor backgrounds compared to children from wealthy backgrounds as they have limited access to comprehensive kidney treatment services. In a similar study, Pais and Wightman (2022) studied the ethical challenges in providing kidney health services to children

and reported that children from poorer areas and families had limited access to sufficient healthcare services compared to children from high-income families hence the high prevalence of chronic kidney disease cases. In an earlier study, Saunders et al. (2020) investigated the influence of neighborhood socioeconomic status and hospitalization due to chronic kidney diseases and found that hospitalization due to chronic kidney disease diagnosis in low-income areas was high compared to high-income areas because of limited access to kidney health services and comprehensive medical insurance covers. The prevalence of chronic kidney disease is directly correlated with poverty, which has contributed to a lack of adequate health services and medical insurance.

Racism. Racism is also a major socioeconomic factor influencing chronic kidney disease prevalence. Mohottige et al. (2021) examined the link between racism and aggression in chronic kidney disease and reported that ethnic minority groups were at a high risk of having chronic kidney disease compared to whites due to limited access to affordable and comprehensive healthcare due to racial discrimination. In a different study, Cerdeña et al. (2021) investigated the effects of racism on the prevalence of chronic kidney disease and established that Black people were more susceptible to suffering from chronic kidney disease than whites due to racism and gene variation. Consistent with Mohottige et al. (2021) and Cerdeña et al. (2021), Eneanya et al. (2022) studied the inequities in healthcare and the prevalence of chronic kidney disease and found that black individuals were more likely to suffer from chronic kidney disease than whites propagated by racism which has led to black individuals having limited access to affordable and conclusive healthcare services. Racism is a socioeconomic factor characterized by the prevalence of chronic kidney disease among blacks, evidenced by limited access to healthcare, education, and housing.

The prevalence of chronic kidney disease is influenced by racism. Through a qualitative study, Crews et al. (2022) studied the influence of racism in managing chronic kidney disease and reported that structural racism has increased chronic kidney disease among black people compared to other races, perpetuated by a lack of medical services and comprehensive treatment

for renal diseases. Lu et al. (2022) examined the influence of racial segregation as a barrier to kidney transplants and found that ethnic minority groups suffering from chronic kidney disease had a lower percentage of transplant referrals compared to whites as a result of racial inequities in healthcare provision. Similar to Lu et al. (2022), Mohottige et al. (2022) investigated the use of race in kidney medicine research and reported that black and ethnic minority groups are most likely to have chronic kidney disease compared to their white peers propagated by lack of access to healthcare services. Racism has increased the cases of chronic kidney diseases by limiting access to healthcare by ethnic minority groups.

Racism has increased the prevalence of the chronic renal disease by restricting ethnic minorities' access to healthcare. Boulware and Mohottige (2021) studied the racial and social inequities affecting kidney care and reported that ethnic minority groups were less likely to receive timely dialysis referrals than non-minorities, thus worsening the prevalence of chronic kidney disease among minority groups. Johansen et al. (2022) examined the variations in the prevalence of end-stage kidney disease among Americans and found that native Hawaiian people and Asians had many cases of chronic kidney disease compared to their peers due to racial biases in the provision of dialysis and kidney transplant referrals. Consistent with Boulware and Mohottige (2021) and Johansen et al. (2022), Rhee et al. (2023) investigated the racial differences in chronic kidney disease risk factors and reported that ethnic minority groups were susceptible to chronic kidney disease compared to majority whites because of racism that impeded access to comprehensive healthcare and medical insurance among the ethnic minority. Racism's restrictions on ethnic minorities' access to healthcare have raised the frequency of chronic kidney illness.

Health Literacy. Health literacy is significant in the self-management of chronic kidney disease by patients. Shah et al. (2021) studied the importance of health literacy in the self-management of chronic kidney disease by conducting a systematic literature review and established that individuals suffering from chronic kidney disease but with prior knowledge of the same managed the adverse effects of chronic kidney disease better than individuals with

limited renal disease information, thus reducing the cases of chronic kidney disease among health literate individuals. Previously, Warsame et al. (2019) examined the effects of low health literacy among kidney transplant patients and found that patients with low literacy levels had little knowledge of the management of chronic kidney disease hence the escalation of cases compared to health-literate individuals who had critical information on comorbidity risk factors and dialysis routines. In contrast, Chu et al. (2020) investigated chronic kidney disease awareness in adults by conducting interviews and reported that individuals with limited health knowledge were at a higher risk of developing chronic kidney disease than health-literate individuals. The prevalence of chronic kidney disease is high among patients with low literacy levels because of limited information concerning the management of renal diseases.

Health literacy is important when managing chronic renal illness on one's own. Iroegbu et al. (2022) examined the awareness of chronic kidney disease and the social determinants of health inequities by conducting a systematic literature review and reported health literacy as among the social determinants of the escalation of chronic kidney disease. Iroegbu et al. (2022) also established that health literacy was important in the early detection of kidney disease hence its management and prevention. Frazier et al. (2022) investigated the decision-making process of older patients with chronic kidney disease by administering questionnaires and found health literacy was important in the decision-making about kidney transplants and dialysis by older patients hence self-management of the diseases compared to limited health literate individuals who were unable to make informed decisions concerning their chronic kidney condition. In a prior study, Taylor et al. (2019) studied the link between health literacy and access to kidney transplants by conducting a qualitative study and found that limited health literacy was associated with reduced access to kidney donor transplants, thus exacerbating chronic kidney disease cases as compared to high literacy individuals who had easy access to deceased and living donor transplants. Patients with chronic kidney disease frequently have low health literacy, which has been correlated with clinical outcomes such as non-adherence to medical treatment.

Health literacy deficiency has escalated chronic kidney disease. Raja et al. (2023) studied the level of knowledge and risk factors associated with chronic illness by using an online survey and established that limited health literacy was a socioeconomic determinant in the increase in cases of chronic kidney diseases because of little information on chronic kidney disease symptoms which prevents its early detection and treatment. Previously, Amaral et al. (2021) investigated the relationship between health literacy and chronic kidney disease and reported that low literacy level was a risk factor in the early stages of chronic kidney disease because of the limited information on the symptoms of the disease hence prevalence in cases among low literacy individuals compared to high health literacy persons. Like Amaral et al. (2021), Lightfoot et al. (2022) examined patient activation as a vital component in managing chronic kidney disease and reported that knowing chronic kidney disease and self-management skills were critical in curbing the rising cases of chronic kidney disease. Health literacy is imperative in managing chronic kidney disease by fully knowing the symptoms and treatment methods.

Education Levels. The level of education is a major socioeconomic factor influencing chronic kidney disease's prevalence. Seng et al. (2020) examined the factors affecting medication adherence among chronic kidney disease patients and established that lower education levels were characterized by poor medical commitment hence the increased prevalence of chronic kidney disease compared to patients with high education levels. In a similar study, Park et al. (2022) studied the inequities in access to organs for transplants and found that education level was key in determining kidney transplants, evidenced by the low number of patients with a low education level receiving kidney transplants compared to patients with a higher level of education. In a prior study, Schrauben et al. (2021) investigated the adoption of technology in the management of chronic kidney disease and reported education levels of patients were critical in adopting technology; thus, patients with low education levels were uncomfortable with using the internet for self-management compared to patients with high education levels. Education level is a significant socioeconomic factor affecting chronic renal disease's prevalence

on account of adherence to medication schedules and the adoption of technology in treating chronic kidney disease by highly educated people.

Education level determines the income, adherence to medical procedures, and access to medical insurance by patients in self-management of chronic kidney disease. El Shamy and Cavanaugh (2020) studied the importance of health education in institutions of learning and reported that health education was vital in curbing cases of chronic kidney disease with early detection of kidney disease among patients with high-level education. In contrast to El Shamy and Cavanaugh (2020), Thio et al. (2020) examined the association between educational levels and the prevalence of chronic kidney disease and found a higher prevalence of chronic kidney disease in individuals with a lower education level compared to highly educated individuals because of limited information on the dietary requirements and non-adherence to medical schedules. In a different study, Tripathy et al. (2020) investigated the association between educational attainment and chronic kidney disease in young adults and reported that low education attainment was highly typified by a higher prevalence of chronic kidney disease compared to higher education attainment evidenced by the improbability of having a higher income and health insurance. Level of income, adherence to prescribed treatments, and availability of health insurance are all influenced by education levels when attempting to treat chronic kidney disease. Figure 7 summarizes the socioeconomic mechanisms described in this section.

The association between race and the prevalence of chronic kidney disease (CKD)

Previous research proves racial disparities in chronic kidney disease (CKD) prevalence. Although chronic kidney disease affects all races, there is increased inequality in chronic kidney disease prevalence resulting in kidney failure among black Americans compared to white Americans (Hounkpatin et al., 2020). A systematic review revealed that black Americans experience an increased rate of kidney failure compared to white Americans. Similar findings to Hounkpatin et al. (2020) were reported by Hsu et al. (2021) in a quantitative study with 1248 participants to investigate Genetic ancestry, race, and kidney function in CKD in the United States. While Hounkpatin et al. (2020) identified hypertension and cardiovascular disease as key causes of

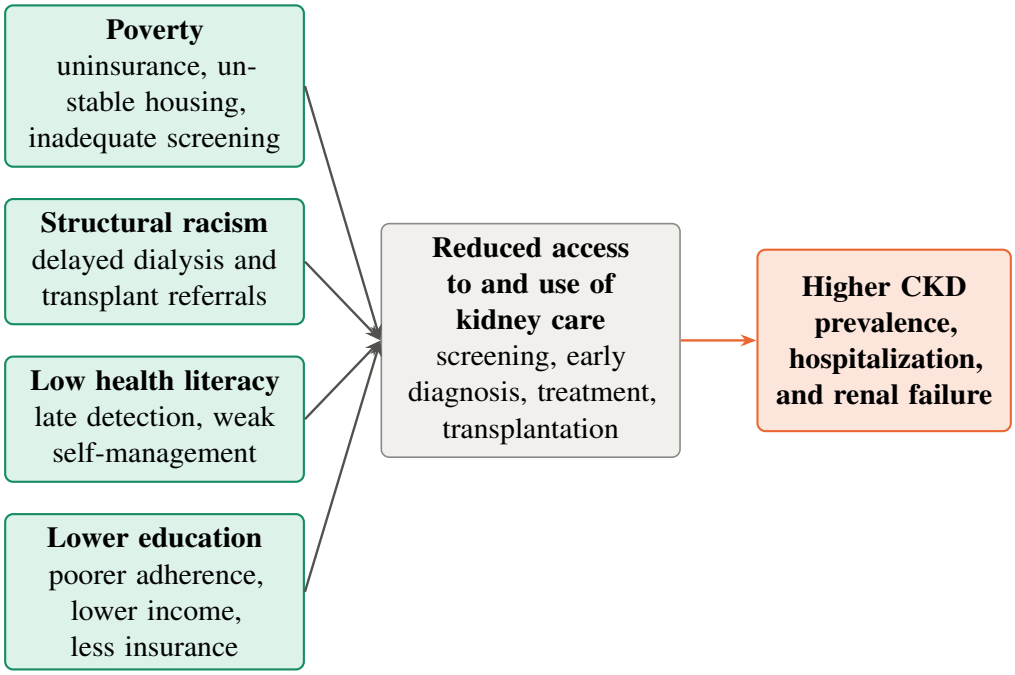


Figure 7

Socioeconomic Determinants of Chronic Kidney Disease Identified in the Literature

Note. Synthesized from the Socioeconomic Status and Prevalence of Chronic Kidney Disease section (e.g., Boulware & Mohottige, 2021; Crews et al., 2022; Morton, 2022; Nguyen et al., 2021; Novick et al., 2022; Saunders et al., 2020; Shah et al., 2021; Thio et al., 2020).

chronic kidney disease among black Americans, Hsu et al. (2021) established a high prevalence of chronic kidney disease because of genetic ancestry. In contrast to Hsu et al. (2021), Vart et al. (2020) revealed that high chronic kidney disease prevalence among minority races could be caused by low socioeconomic status restricting them from accessing quality healthcare services. Although the articles reviewed highlighted different causes of chronic kidney disease, the investigators identified a high prevalence of CKD among minority races in the United States.

Past research provides empirical evidence that chronic kidney disease has been prevalent among minority races compared to white races. In a quantitative study, Kovesdy (2022) investigated the epidemiology of chronic kidney disease across different races in the United States. The findings demonstrated that African Americans are more at risk of kidney failure than other races in the United States (Kovesdy, 2022). Levey et al. (2020) also established that chronic kidney disease is prevalent among minority groups such as black Americans. Although the researcher identified high CKD prevalence among minority races, kidney failure has been the cause of death across all races in the United States (Levey et al., 2020). Levey et al. (2020) suggested enhanced efforts for better prevention and treatment of kidney diseases to reduce the high prevalence of CKD among minority groups. Purnell et al. (2020) reported consistent findings of Levey et al. (2020), establishing that compared to whites, minority races have a high rate of chronic kidney disease prevalence because of disparity in medical access. Given the findings, it is evident that chronic kidney disease is highly prevalent among minority races compared to white races.

Healthcare disparities among the races may have a significant impact prevalence of chronic kidney disease in the United States. Bragg-Gresham et al. (2021) examined the prevalence of chronic kidney disease among black individuals in the United States and established the need for equality in healthcare access. However, a high level of racial segregation makes it difficult for blacks to access quality medication for kidney failure resulting in an increased prevalence of chronic kidney disease among the minority races as opposed to the white races (Bragg-Gresham et al., 2021). Further, previous research highlighted that racial and ethnic

disparities had influenced chronic kidney disease among individuals in accessing essential services such as medical care (Eneanya et al., 2022). Whereas chronic kidney disease has been rampant across all racial groups, racial disparity, and access to healthcare services for better treatment have been the key reason for the high prevalence of chronic kidney disease in minority groups. Studies have shown that blacks and Hispanics have higher CKD prevalence levels than whites (Eneanya et al., 2022; Bragg-Gresham et al., 2021). The articles reviewed thus far have indicated a high prevalence of chronic kidney disease among non-white people in the United States.

Being common in minority groups, sickle cell disease has been linked to an increased chronic kidney disease prevalence. Olaniran et al. (2019) conceptualized cardiovascular results in black Americans with sickle cell trait and kidney disease in the United States. Although sickle cell disease has been associated with a high prevalence of chronic kidney disease and cardiovascular risks in the general population, there has been a high chronic kidney disease among African Americans (Olaniran et al., 2019). Corresponding findings to Olaniran et al. (2019) were established by Murphy et al. (2020) on the prevalence of chronic kidney disease among African Americans with hypertension and kidney failure in the United States. The findings revealed that a lack of awareness of chronic kidney disease led to an increased rate of chronic kidney disease prevalence among African Americans in the United States (Murphy et al., 2020). Though Murphy et al. (2020) reported similar findings to Olaniran et al. (2019), indicating a high prevalence of chronic kidney disease among minority races, the researcher conducted the study in one location, limiting the generalizability of findings. Thus far, studies analyzed demonstrate the high prevalence of chronic kidney disease among minority groups compared to majority races.

Chronic kidney disease has been observed in minority races than white races. A qualitative study with 23 participants by Lissanu et al. (2019) revealed that although minority and majority races are affected by the increased prevalence of chronic kidney disease, minority races are the most affected. In this regard, Lissanu et al. (2019) recommended creating awareness

regarding the severity of CKD and how to manage CKD among patients. Consistent results of Lissanu et al. (2019) were established by Ku et al. (2020) in a quantitative study with 678 participants to conceptualize race and mortality in CKD and dialysis in the United States. The authors demonstrated a high prevalence of chronic kidney disease and the resultant mortality rate among black Americans in the United States (Ku et al., 2020). In contrast to Ku et al. (2020) findings, Hall et al. (2019) focused on functional and structural social support with chronic disease and highlighted that minority races with low functional and social support were associated with a lower likelihood of CKD prevalence. Though some researchers cited low social support causing increased CKD prevalence among blacks, other articles agree that minority races are vulnerable to CKD.

Earlier research indicates the relationship between racial healthcare disparities and chronic kidney disease prevalence. Limited access to healthcare among minority races could be the for the increased prevalence of chronic kidney disease (Minhas et al., 2021). Black women are the most affected by racial disparity in healthcare because of their risk for acute cardiovascular complications during delivery, which may lead to high chronic kidney disease prevalence among minority races compared to white races (Minhas et al., 2021). Although Minhas et al. (2021) focused on black women and the risk for acute cardiovascular complications contributing to increased chronic kidney disease, Jacobson et al. (2020) opined that more minority adolescents are enrolled in chronic kidney disease treatment centers compared to white adolescents with efficient healthcare services for chronic disease prevention and treatment. In a quantitative study, Miller et al. (2022) investigated national kidney foundation and kidney treatment prevention and racial disparity in the United States and established disparity in national kidney prevention and treatment across different races. While the articles reviewed indicate differing results regarding the prevalence of chronic kidney disease, minority races are more affected by chronic kidney disease than the majority races.

There are black-white inequalities in chronic kidney disease prevalence in the United States. To better understand the impact of healthcare disparities and chronic kidney disease

prevalence, Iantorno et al. (2019) examined race, cardiovascular disease, and the prevalence of chronic kidney disease. The researchers established that the black population experiences healthcare disparities in treating conditions such as diabetes, coronary artery disease, and acute myocardial infarction, resulting in a higher prevalence of chronic kidney disease than the white population (Iantorno et al., 2019). A quantitative study with 359 participants conducted by Kwan et al. (2020) demonstrated that although the researcher reported consistent findings to Iantorno et al. (2019) by revealing that minority races were most affected by chronic kidney disease than whites, Kwan et al. (2020) focused on urine metabolites to get insight into the chronic kidney disease progression. However, Akhabue et al. (2023) examined cardiovascular hospitalization trends across race and ethnic groups and found that chronic kidney disease hospitalization rate was declining for non-Hispanic and black groups than whites. The articles reviewed reveal that the non-white race is more affected by the prevalence of chronic kidney disease.

Ethnic and racial disparities in diabetes-related chronic conditions have been associated with chronic kidney disease in the United States. Clements et al. (2020) conducted a quantitative study with 443,932 Medicare beneficiaries to investigate disproportionate diabetes-linked multiple chronic conditions and mortality in the United States to assert the claim. The results demonstrated that compared to the white race, the black race had increased odds of being most affected by CKD prevalence, resulting in an increased risk of mortality and kidney failures (Clements et al., 2020). By reassessing the inclusion of race in diagnosing kidney disease, Delgado et al. (2021) cited that the lack of diagnosis of kidney disease among minorities due to inequality in healthcare led to increased chronic kidney disease prevalence. Ku et al. (2020) conducted racial and ethnic disparities in kidney transplant access and its impact on chronic kidney disease prevalence. Limited access to medication and kidney transplants contributed to a high prevalence of CKD among minority races compared to white ones (Ku et al., 2020). Studies discussed thus far have provided significant insight into the reason for the disparity in chronic kidney disease prevalence across races.

Despite some whites facing a risk of death because of renal and kidney failure, chronic kidney disease has been more prevalent among minority races. Previous research suggests that compared to whites, blacks experience a higher risk of morbidity and mortality related to chronic kidney disease (Yu et al., 2019). Yu et al. (2019) highlighted that black Americans are more at risk of developing kidney failure than whites in the United States. Murphy et al. (2019) corroborated Yu et al. (2019) and revealed that blacks are more likely to die of chronic kidney disease than whites. The reason for such discrepancies is the inequality in healthcare access among the minority race as opposed to whites with effective medical care for renal disease management (Murphy et al., 2019). Though Murphy et al. (2019) and Yu et al. (2019) agreed that healthcare disparity contributes to the increased prevalence of chronic kidney disease, Thurlow et al. (2021) identified low socioeconomic status and limited medical access contribute to a high prevalence of the chronic disease among minority races compared to white. The articles reviewed thus far have shown non-white races being the most affected with a high prevalence of chronic kidney disease.

Racial difference is another key factor that influences the prevalence of CKD. Several studies conducted in the United States have revealed the significant influence of racial differences on the prevalence of CKD (Boynton et al., 2022; Nguyen et al., 2021). More specifically, researchers such as Norris and Beech (2021) have found that the prevalence of CKD among minorities is higher than among the dominant races. Other researchers have also reported similar results linking racial differences to an increase in CKD prevalence; some of the reasons identified to contribute to the racial differences in CKD prevalence are socioeconomic status and geographical locations (Morton, 2022; Novick et al., 2022). Low socioeconomic status among minorities limits their access to medical services, such as screening that could have helped them detect CKD in the initial stages (Purnell et al., 2019). Equally, low socioeconomic status among minorities makes it difficult for them to access proper diets or healthy lifestyles that could improve their overall well-being or mitigate the risk factors for CKD. While some studies have researched the differences in CKD prevalence across races, most studies were old

(Iorember & Bamgbola, 2022). CKD is a dynamic condition whose comorbidity and progression keep evolving, making it difficult to focus on longitudinal studies (Saunders et al., 2020). It is therefore important to conduct the current study using up-to-date data to establish and address the existing inconsistencies in CKD prevalence across races. This is important as it will provide an opportunity to better understand the trends and changes in CKD prevalence.

Conclusion

Several reasons warrant this study despite extensive literature on the subject. First, there is still a paucity of data on the relationship between the area of residence SES and severity of established CKD at presentation to a renal service in the United Kingdom. Second, the previous work on this topic concentrated mainly on patients with ESRD. Third, differences across regions and countries in the population mix, access, and quality of health care, and even lifestyle may have a different impact on the relationship between SES and CKD. The underlying hypothesis in this study, therefore, was whether characteristics of an area of residence, in terms of measures of SES, have an impact on the level of kidney function of patients with CKD at presentation. Therefore, this study investigates the relationship between renal failure and length of stay, total charges, and their association with age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States.

CHAPTER III: METHODOLOGY

Subjects

The purpose of this study is to investigate the predictors of renal failure based on length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. This study's data collection instrument is archival data, specifically the patient records of the selected patients in the United States. This study will collect and analyze data regarding the predictors of renal failure from the Nationwide Inpatient Sample (NIS) data set for the years 2010, 2011, and 2012. The problem is that it remains unclear the associations of renal failure with the length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities among inpatients in the United States. Correspondingly, the following research questions with hypotheses will be assessed:

RQ1: Is there a relationship between renal failure and length of stay?
H_0: There is no significant relationship between renal failure and length of stay.
H_1: There is a significant relationship between renal failure and length of stay.

RQ2: Is there a relationship between renal failure and total charges?
H_0: There is no significant relationship between renal failure and total charges.
H_1: There is a significant relationship between renal failure and total charges.

RQ3: Is there a relationship between renal failure and age?
H_0: There is no significant relationship between renal failure and age.
H_1: There is a significant relationship between renal failure and age.

RQ4: Is there a relationship between renal failure and gender?
H_0: There is no significant relationship between renal failure and gender.
H_1: There is a significant relationship between renal failure and gender.

RQ5: Is there a relationship between renal failure and race?
H_0: There is no significant relationship between renal failure and race.
H_1: There is a significant relationship between renal failure and race.

RQ6: Is there a relationship between renal failure and socioeconomic status (SES)?

H_0: There is no significant relationship between renal failure and socioeconomic status (SES).

H_1: There is a significant relationship between renal failure and socioeconomic status (SES).

RQ7: Is there a relationship between renal failure and the number of procedures?

H_0: There is no significant relationship between renal failure and the number of procedures.

H_1: There is a significant relationship between renal failure and the number of procedures.

RQ8: Is there a relationship between renal failure and the number of diagnoses?

H_0: There is no significant relationship between renal failure and the number of diagnoses.

H_1: There is a significant relationship between renal failure and the number of diagnoses.

RQ9: Is there a relationship between renal failure and comorbidities?

H_0: There is no significant relationship between renal failure and comorbidities.

H_1: There is a significant relationship between renal failure and comorbidities.

RQ10: Do length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively predict renal failure?

H_0: Length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively do not significantly predict renal failure.

H_1: Length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively significantly predict renal failure.

This study's research methodology and design is a quantitative correlational design study. A quantitative research methodology is based on an objectivity paradigm, uses numerical data to enable statistical analyses, and helps minimize biases (Bowers, 2017). Statistical, mathematical, and numerical analyses of data obtained from surveys and questionnaires and using computational methods to manipulate statistical data that has already been collected are examples of quantitative research measures (Bowers, 2017).

A non-experimental quantitative methodology with a correlational design is most appropriate for specific reasons. First, the study includes numerical data analyzed to test hypotheses (McCusker & Gunaydin, 2015). Second, choosing a nonexperimental quantitative method with a correlational design ensures research objectivity as the researcher is separated from the research participants (McCusker & Gunaydin, 2015). Third, there is no manipulation of independent

variables; thus, this study is a nonexperimental quantitative method with a correlational design (McCusker & Gunaydin, 2015). Additionally, a nonexperimental quantitative method with a correlational design is the correct design for the current study because the objective is to identify and evaluate the relationship between the dependent and independent variables.

In a correlational research design, two variables are measured without the researcher controlling either (McCusker & Gunaydin, 2015). It is appropriate to conduct this study using a quantitative correlational design because its goal is to identify and assess the relationship between renal failure, the dependent variable, and the nine independent variables (length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities).

A qualitative approach is inappropriate because the study does not focus on exploring a phenomenon or establishing a theory, model, or definition (Allwood, 2012). Due to the nature of the research questions posed, binary logistic regression is the best fit for data analysis for this study. Binary logistic regression analysis is used to predict a dichotomous dependent variable, renal failure (yes or no) in this case, based on independent variables, length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities (Field, 2018). Additionally, binary logistic regression analysis also determines the overall fit and the relative contribution of each predictor to the total variance explained (Field, 2018). In binary logistic regression, covariates may be added to the model to control the effects they may have.

The complete research design and analytic sequence is depicted in Figure 8.

Instrument

This study's data collection instrument will be archival data of the selected inpatients in the United States. The researcher targets the data from Nationwide Inpatient Sample (NIS) datasets for 2010, 2011, and 2012. Table 1 provides the variable names, labels, and operationalization of each variable of this study. Binary logistic regression will assess the relationship between the dependent variable, renal failure, and the independent variables of the length of stay, total

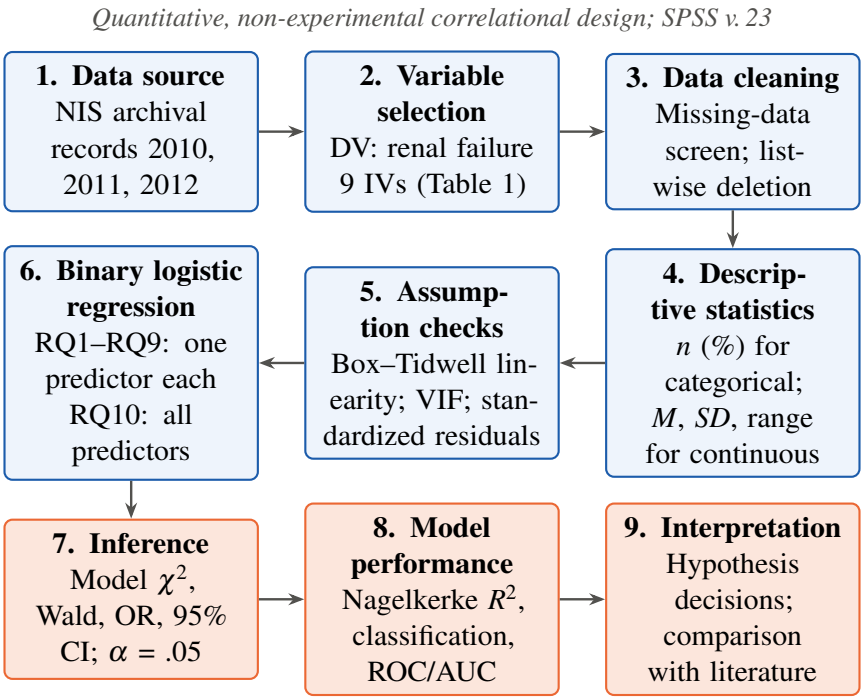


Figure 8

Research Design and Analytic Workflow

Note. Prepared from the procedures described in Chapter III. DV = dependent variable; IV = independent variable; VIF = variance inflation factor; OR = odds ratio; CI = confidence interval; ROC = receiver operating characteristic; AUC = area under the curve.

charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities.

Table 1
Operationalization of Study Variables

Variable name	Label	Measurement type	Operationalization
CM_RENLFAIL	Renal failure (dependent variable)	Nominal, dichotomous	1 = present, 0 = not present
AGE	Age in years at admission	Interval	Age in years
CM_AIDS	AHRQ comorbidity measure for ICD-9-CM codes: acquired immune deficiency syndrome	Nominal, dichotomous	1 = present, 0 = not present
CM_ALCOHOL	AHRQ comorbidity measure for ICD-9-CM codes: alcohol abuse	Nominal, dichotomous	1 = present, 0 = not present
CM_ANEMDEF	AHRQ comorbidity measure for ICD-9-CM codes: deficiency anemias	Nominal, dichotomous	1 = present, 0 = not present
CM_ARTH	AHRQ comorbidity measure for ICD-9-CM codes: rheumatoid arthritis/collagen vascular diseases	Nominal, dichotomous	1 = present, 0 = not present
CM_BLDLOSS	AHRQ comorbidity measure for ICD-9-CM codes: chronic blood loss anemia	Nominal, dichotomous	1 = present, 0 = not present
CM_CHF	AHRQ comorbidity measure for ICD-9-CM codes: congestive heart failure	Nominal, dichotomous	1 = present, 0 = not present
CM_CHRNLUNG	AHRQ comorbidity measure for ICD-9-CM codes: chronic pulmonary disease	Nominal, dichotomous	1 = present, 0 = not present
CM_COAG	AHRQ comorbidity measure for ICD-9-CM codes: coagulopathy	Nominal, dichotomous	1 = present, 0 = not present
CM_DEPRESS	AHRQ comorbidity measure for ICD-9-CM codes: depression	Nominal, dichotomous	1 = present, 0 = not present
CM_DM	AHRQ comorbidity measure for ICD-9-CM codes: diabetes, uncomplicated	Nominal, dichotomous	1 = present, 0 = not present
CM_DMCX	AHRQ comorbidity measure for ICD-9-CM codes: diabetes with chronic complications	Nominal, dichotomous	1 = present, 0 = not present
CM_DRUG	AHRQ comorbidity measure for ICD-9-CM codes: drug abuse	Nominal, dichotomous	1 = present, 0 = not present

continued on next page

Table 1 (continued)

Variable name	Label	Measurement type	Operationalization
CM_HTN_C	AHRQ comorbidity measure for ICD-9-CM codes: hypertension (combine uncomplicated and complicated)	Nominal, dichotomous	1 = present, 0 = not present
CM_HYPOTHY	AHRQ comorbidity measure for ICD-9-CM codes: hypothyroidism	Nominal, dichotomous	1 = present, 0 = not present
CM_LIVER	AHRQ comorbidity measure for ICD-9-CM codes: liver disease	Nominal, dichotomous	1 = present, 0 = not present
CM_LYMPH	AHRQ comorbidity measure for ICD-9-CM codes: lymphoma	Nominal, dichotomous	1 = present, 0 = not present
CM_LYTES	AHRQ comorbidity measure for ICD-9-CM codes: fluid and electrolyte disorders	Nominal, dichotomous	1 = present, 0 = not present
CM_METS	AHRQ comorbidity measure for ICD-9-CM codes: metastatic cancer	Nominal, dichotomous	1 = present, 0 = not present
CM_NEURO	AHRQ comorbidity measure for ICD-9-CM codes: other neurological disorders	Nominal, dichotomous	1 = present, 0 = not present
CM_OBESE	AHRQ comorbidity measure for ICD-9-CM codes: obesity	Nominal, dichotomous	1 = present, 0 = not present
CM_PARA	AHRQ comorbidity measure for ICD-9-CM codes: paralysis	Nominal, dichotomous	1 = present, 0 = not present
CM_PERIVASC	AHRQ comorbidity measure for ICD-9-CM codes: peripheral vascular disorders	Nominal, dichotomous	1 = present, 0 = not present
CM_PSYCH	AHRQ comorbidity measure for ICD-9-CM codes: psychoses	Nominal, dichotomous	1 = present, 0 = not present
CM_PULMCIRC	AHRQ comorbidity measure for ICD-9-CM codes: pulmonary circulation disorders	Nominal, dichotomous	1 = present, 0 = not present
CM_TUMOR	AHRQ comorbidity measure for ICD-9-CM codes: solid tumor without metastasis	Nominal, dichotomous	1 = present, 0 = not present
CM_ULCER	AHRQ comorbidity measure for ICD-9-CM codes: peptic ulcer disease excluding bleeding	Nominal, dichotomous	1 = present, 0 = not present
CM_VALVE	AHRQ comorbidity measure for ICD-9-CM codes: valvular disease	Nominal, dichotomous	1 = present, 0 = not present
CM_WGHTLOSS	AHRQ comorbidity measure for ICD-9-CM codes: weight loss	Nominal, dichotomous	1 = present, 0 = not present

continued on next page

Table 1 (continued)

Variable name	Label	Measurement type	Operationalization
FEMALE	Indicator of sex	Nominal, dichotomous	1 = female, 0 = male
LOS	Length of stay in days	Interval	Number of days
NDX	Number of ICD-9-CM diagnoses on this discharge	Interval	Number of diagnoses
NPR	Number of ICD-9-CM procedures on this discharge	Interval	Number of procedures
RACE	Race/ethnicity of the patient	Nominal	1 = White; 2 = Black; 3 = Hispanic; 4 = Asian or Pacific Islander; 5 = Native American; 6 = Other
TOTCHG	Total charges (in dollars)	Interval	Total charges (in dollars)
ZIPINC	Median household income for patient's ZIP Code (based on 1999 demographics)	Nominal	1 = $1–24,999; 2 = $25,000–34,999; 3 = $35,000–44,999; 4 = $45,000 or more

Note. AHRQ = Agency for Healthcare Research and Quality; ICD-9-CM = International Classification of Diseases, Ninth Revision, Clinical Modification. The 27 AHRQ comorbidity indicators other than renal failure were summed to form the number-of-comorbidities predictor (Table 7).

Reliability

Reliability is a measure's consistency (Twycross & Shields, 2004). The correlational research design has the advantage that neither variable is altered, and it is irrelevant how or where the variables are measured (McCusker & Gunaydin, 2015). As a result, the study is trustworthy in terms of its design. Additionally, the instrument or source for data collection will be archival data. Archival data are the data the research site has gathered, making data collection and measurement straightforward. As a result, this study is reliable in terms of the instruments used.

Validity

Validity consists of two types: external and internal validity. External validity refers to the degree to which the study results can be generalized to the population. Studies utilizing convenience sampling present challenges to external validity (Etikan, 2016). Studies that involve purposive

samples may have issues with the generalizability of the study findings to broader populations of interest (Etikan, 2016).

Internal validity refers to the validity of the findings within the research study. Testing hypotheses can threaten the validity of interpretation for quantitative researchers, and quantitative research may involve rejecting null hypotheses or failing to reject null hypotheses (Martin & Bridgmon, 2012). Consequently, threats to conclusive findings occur when quantitative researchers encounter a Type I error, which involves rejecting a valid null hypothesis (Ibrahim, Ghani, & Embat, 2013).

Procedures

The goals of the current study will be met using data from the Nationwide Inpatient Sample (NIS). This data will be downloaded with permission, and their use will be authorized. Most clinicians and researchers frequently used the NIS data in their studies, particularly those that focused on the impact of patient information on the length of stay, overall costs, and mortality. The demographics, hospital characteristics, admission types and years, and comorbidities make up most of the patient data in the NIS database.

Because an existing dataset will be used, this study will not require informed consent procedures, and data will be retrieved from the publicly available NIS dataset. Because participants are not identifiable in the data, no special precautions will be required to safeguard the anonymity of participants.

A priori power analysis was conducted using G*Power to determine the required minimum sample size for the study. Four factors were considered in the power analysis: significance level, effect size, the power of the test, and statistical technique. The significance level, also known as Type I error, refers to the chance of rejecting a null hypothesis if it is true (Haas, 2012). Most quantitative studies use a 95% confidence level because it adequately provides enough statistical evidence of a test (Creswell & Poth, 2017). The effect size refers to the estimated measurement of the relationship between the considered variables (Cohen, 1988). Cohen (1988) categorizes effect size into small, medium, and large. Berger, Bayarri, and Pericchi (2013)

purported that a medium effect size is better as it strikes a balance between being too strict (small) and too lenient (large). The power of a test refers to the probability of correctly rejecting a null hypothesis (Sullivan & Feinn, 2012). In most quantitative studies, an 80% power is usually used (Sullivan & Feinn, 2012). The statistical test to be used for this study is binary logistic regression. The calculation of a minimum sample size for logistic regression requires previous knowledge, such as the expected odds ratio (effect size), the proportion of observations in either group of the dependent variable, and the distribution of each independent variable (Faul et al., 2009). If these are unknown, it is best to use an estimate to determine the appropriate sample size. Using G*Power, the minimum sample size was computed by utilizing a medium effect size of OR = 2.47, based on the categorization of effect sizes by Chinn (2000), who categorized odds ratio into small (OR = 1.44), medium (OR = 2.47), and large (OR = 4.25). To conduct binary logistic regression to detect a medium effect size of OR = 2.47, at the 5% significance level, with 80% power, a minimum sample size of at least 72 (Figure 9). The NIS dataset consists of data from over seven million hospital stays at the patient level; thus, the sample size required for the study is met.

To verify the a priori calculation and show how power depends on sample size and effect size, a Monte Carlo simulation (1,500 replications per sample size) was run with the same G*Power inputs: a standard-normal predictor, a baseline event probability of .20, and a two-sided Wald test at $\alpha = .05$. Figure 10 shows that with the medium effect size used for the study (OR = 2.47), simulated power reaches approximately 80% at $N = 72$, which agrees with the G*Power result. It also shows that the analytic sample of 2,313,779 discharges gives power close to 1.0 for even small effects, which is why statistical significance in Chapter IV should be read together with the size of each odds ratio.

Data Analysis

The resulting quantitative data analysis will be conducted using the statistical software suite Statistical Package for the Social Sciences (SPSS) version 23. The data will be cleaned by examining the dataset for missing data (Field, 2018). If a value is missing, the entire case will

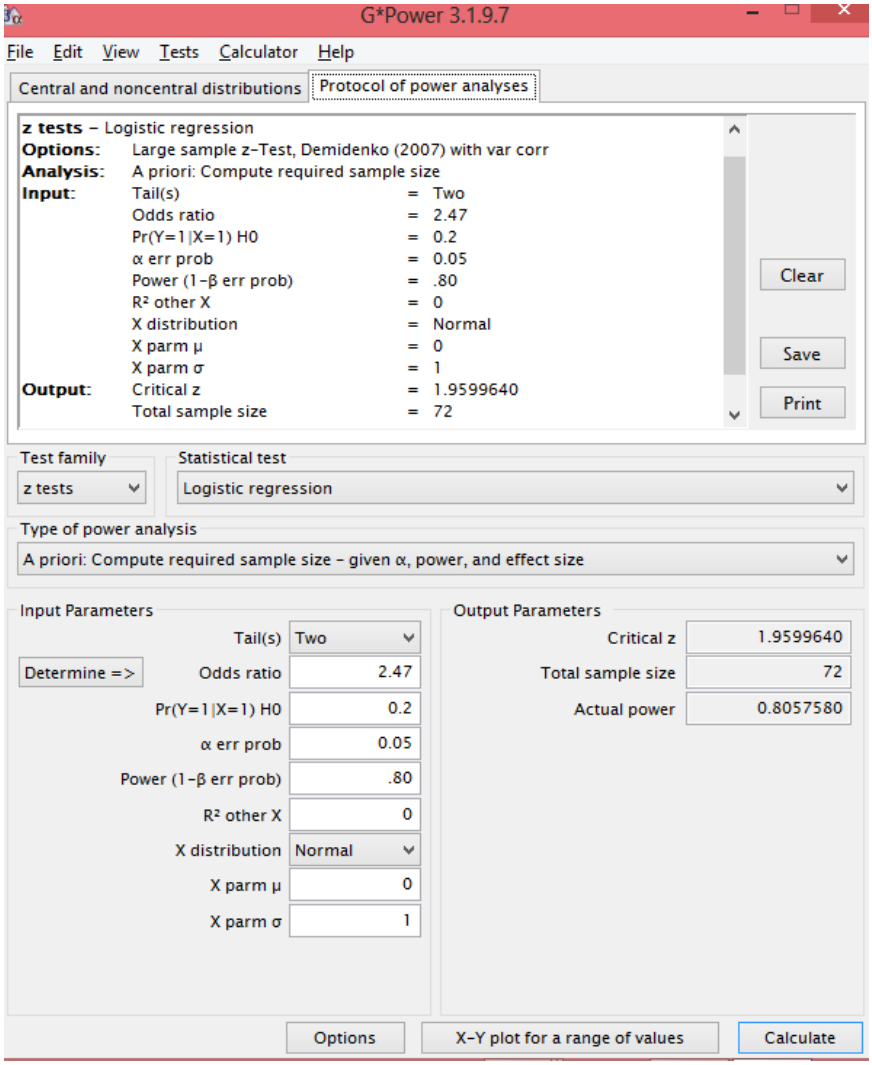


Figure 9

*G*Power Sample Estimation for Binary Logistic Regression*

Note. Protocol of the a priori power analysis (G*Power 3.1.9.7; large-sample z test, Demidenko, 2007, with variance correction): two-tailed, OR = 2.47, $\Pr(Y = 1 \mid X = 1)$ under $H_0 = .20$, $\alpha = .05$, power = .80, $X \sim N(0, 1)$. Required total sample size = 72 (actual power = .806).

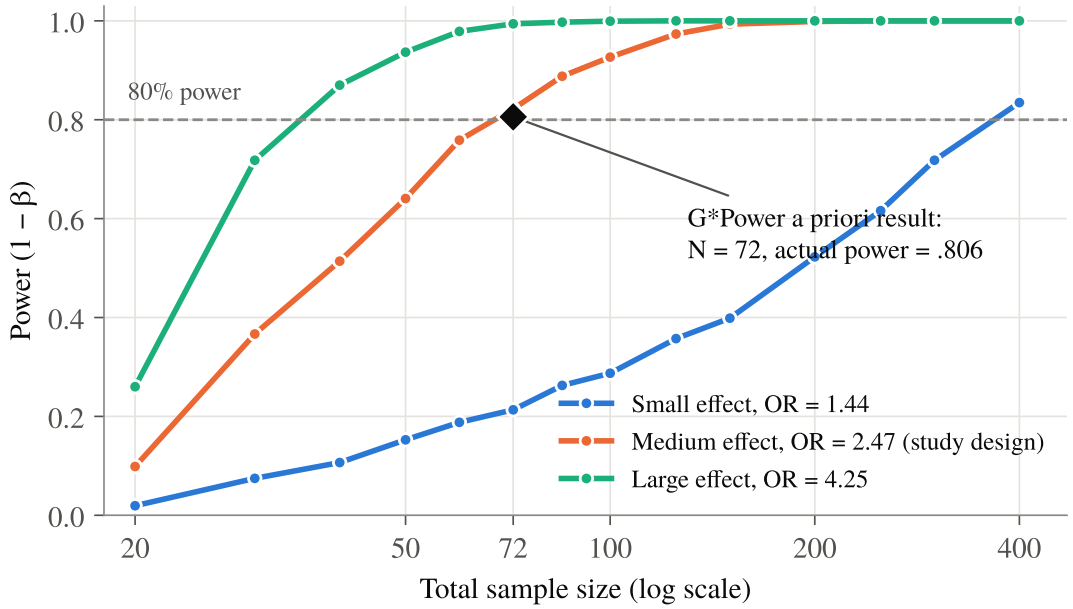


Figure 10

Statistical Power of the Wald Test for Binary Logistic Regression by Sample Size and Effect Size

Note. Monte Carlo estimates (1,500 replications per point; seed 2023) using the parameters of Figure 9. Effect-size benchmarks follow Chinn (2000): small OR = 1.44, medium OR = 2.47, large OR = 4.25. The diamond marks the G*Power solution.

be removed from the analysis (listwise deletion). In listwise deletion, a case is dropped from the analysis because it has a missing value in at least one of the specified variables. The analysis is only run on cases with a complete data set.

Descriptive statistics of the data for the predictor and dependent variables will be reported. Frequency and percentages summary will be obtained for categorical variables, while the measure of central tendencies of means and standard deviations and minimum and maximum values will be conducted for continuous demographic variables, such as age.

In order to address the research questions, binary logistic regression will be conducted to test the following model:

$$\begin{aligned}\text{logit}(\text{Renal failure}) = {} & \beta_0 + \beta_1\,\text{Length of stay} + \beta_2\,\text{Total charges} + \beta_3\,\text{Age} + \beta_4\,\text{Gender} \\ & + \beta_5\,\text{Race} + \beta_6\,\text{SES} + \beta_7\,\text{Number of procedures} \\ & + \beta_8\,\text{Number of diagnoses} + \beta_9\,\text{Comorbidities}\end{aligned} \tag{1}$$

where $\text{logit}(p) = \ln[p/(1-p)]$ and p is the probability that renal failure is present. Each odds ratio is obtained as $\text{OR} = e^{\beta}$.

The comorbidities variable consists of 28 separate dichotomous variables, as presented in Table 1. Prior to conducting binary logistic regression, some assumptions must be met. These include linearity between the continuous independent variables and the logit transformation of the dependent variable, the absence of multicollinearity, and the absence of significant outliers (Laerd Statistics, 2017). Linearity will be tested using the Box-Tidwell procedure (Laerd Statistics, 2017). Multicollinearity will be tested by calculating variance inflation factors (VIF), and any VIF over 10 will be considered evidence of multicollinearity (Laerd Statistics, 2017). Standardized residuals will be calculated to test for outliers, and any standardized residual beyond three standard deviations ($|z| > 3.0$) will be considered an outlier (Laerd Statistics, 2017). Once the assumptions are tested, the independent and dependent variables will be entered into SPSS. Significance will be assessed at the 5% level; thus, any p-value less than or equal to 0.05 will be deemed significant.

Limitations

A limitation is a boundary imposed on the study by its nature. A quantitative analysis assumes no interference or manipulation of data for the study (Field, 2018). The self-reporting data by individuals is one limitation of the study. This self-reporting data may limit the reliability of the data tool if the participants are not truthful regarding the information provided, such as annual income. It is assumed, however, that the data is accurate and reliable. Another limitation of the study is the use of convenience sampling which limits the generalizability of study findings relative to probabilistic or random sampling techniques. Additionally, correlational design cannot deduce any cause-and-effect relationships between the study variables, as independent variables will not be manipulated.

Summary

The purpose of this study is to investigate the predictors of renal failure based on length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. This study will collect and analyze data regarding the predictors of renal failure from the Nationwide Inpatient Sample (NIS) data set for the years 2010, 2011, and 2012. The ten research questions and hypotheses stated at the beginning of this chapter will be addressed by employing binary logistic regression.

This chapter comprehensively describes the quantitative correlational research design used for this study. The results and findings from the data analysis will be presented in Chapter 4, along with the tables and graphics providing the descriptive results and inferences regarding the underlying connection between the study variables. The interpretations of the findings are provided in Chapter 5, along with the study's limitations, recommendations for future studies, and implications for positive social change.

CHAPTER IV: RESULTS

The purpose of this study was to investigate the predictors of renal failure based on length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. This study's data collection instrument was archival data, specifically the patient records of the selected patients in the United States. This study collected and analyzed data regarding the predictors of renal failure from the Nationwide Inpatient Sample (NIS) data set for the years 2010, 2011, and 2012. The ten research questions and hypotheses stated in Chapters I and III (RQ1–RQ10; see also Figure 1) were addressed.

The following is a discussion of the study's population and sample and a demographic description of the sample. Demographic descriptions included frequencies and percentages for categorical (nominal) variables and mean and standard deviations measured at the interval level of measurement. Also presented are the testing of parametric assumptions for the statistical analysis and the results of statistical testing. This chapter concludes with a discussion of the results of this study.

Data Collection

Data were retrieved from the Nationwide Inpatient Sample (NIS) data set for 2010, 2011, and 2012. Specifically, the variables included were renal failure, length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities. The utilized data set consisted of N = 2,313,779 cases which included 975480 (42.2%) males and 1334806 (57.7%) females. There were 3493 (0.2%) cases in which no sex was specified (Table 2).

The mean age was M = 48.87 (SD = 27.61). Regarding race, most were White, 1385592 (59.9%). This was followed by Black, 322087 (13.9%); Hispanic, 253682 (11.0%); Asian, 54186 (2.3%); some other race, 71816 (3.1%); and Native American, 14692 (0.6%). Table 3 provides this information.

Table 2
Sex

	Frequency	Percent
Males	975,480	42.2
Females	1,334,806	57.7
Missing	3,493	0.2
Total	2,313,779	100.0

Table 3
Race

	Frequency	Percent
White	1,385,592	59.9
Black	322,087	13.9
Hispanic	253,682	11.0
Asian or Pacific Islander	54,186	2.3
Native American	14,692	0.6
Other	71,816	3.1
Missing	211,724	9.2
Total	2,313,779	100.0

Socioeconomic status (SES) was measured using rankings from poorest to wealthiest and then grouping them into four income quartiles (1 being poorest and 4 being wealthiest), each containing approximately 25% of the population. This information is provided in Table 4.

Table 4

Median Household Income National Quartile for Patient ZIP Code

	Frequency	Percent
1st quartile (poorest)	672,541	29.1
2nd quartile	566,608	24.5
3rd quartile	552,549	23.9
4th quartile (wealthiest)	469,455	20.3
Missing	52,626	2.3
Total	2,313,779	100.0

Figure 11 displays the demographic and socioeconomic composition of the sample summarized in Tables 2–4. Female discharges outnumbered male discharges, three in five patients were White, and the poorest income quartile was the largest group. The number of

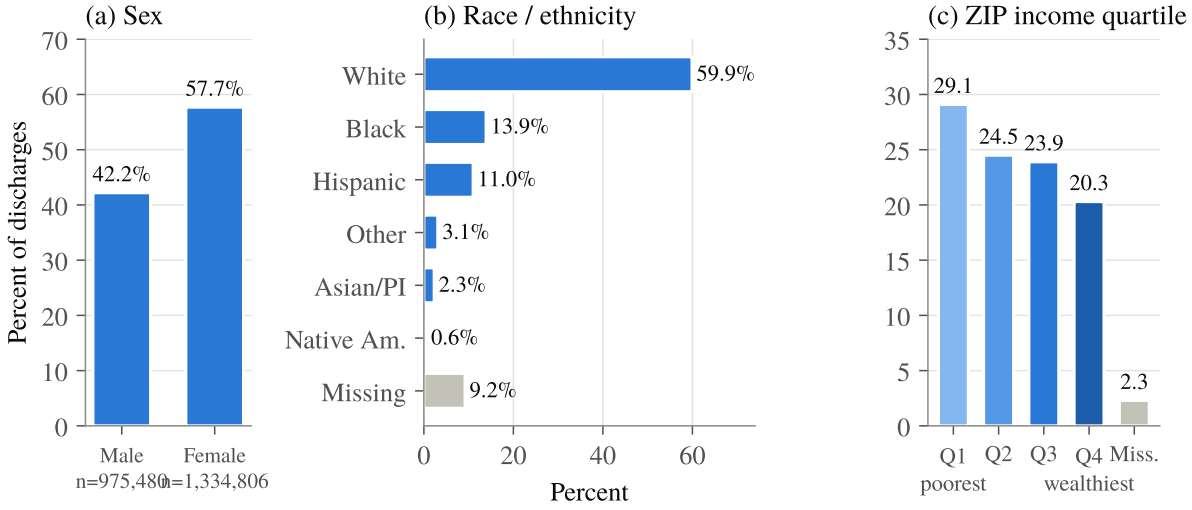


Figure 11

Demographic and Socioeconomic Profile of the Study Sample

Note. Percentages of all N = 2,313,779 discharges (Tables 2–4). Panel (a) omits the 0.2% of records with sex unspecified. Asian/PI = Asian or Pacific Islander.

procedures ranged from 0 to 39 (M = 1.62, SD = 2.11); the number of diagnoses ranged from 0

to 59 (M = 8.61, SD = 5.73); total charged ranged from 101 to 4831287 (M = 34981.09, SD = 63555.55); and length of stay ranged from 0 to 365 days (M = 4.57, SD = 6.72). Table 5 provides this information.

Table 5
Descriptive Statistics for the Number of Procedures, Diagnoses, Total Charges, and Length of Stay

Variable	*N*	Minimum	Maximum	*M*	*SD*
Number of procedures	2,313,779	0	39	1.62	2.113
Number of diagnoses	2,313,779	0	59	8.61	5.726
Total charges ($)	2,266,240	101	4,831,287	34,981.09	63,555.547
Length of stay (days)	2,313,188	0	365	4.57	6.721

The distributions in Table 5 and Table 7 are strongly right-skewed: for length of stay, total charges and number of procedures the standard deviation exceeds the mean, and the maxima lie far above the means. Figure 12 places the mean and one standard deviation of each continuous predictor side by side, together with its observed range. Regarding comorbidities, there were 28 in the data set. Renal failure was of the comorbidities in the data file coded as 0 for no or 1 for yes, which served as the dependent variable. There were 220821 (9.5%) occurrences of renal failure and 2092958 (90.5%) non-occurrences (Table 6).

Table 6
AHRQ Comorbidity Measure: Renal Failure

	Frequency	Percent
Absent	2,092,958	90.5
Present	220,821	9.5
Total	2,313,779	100.0

Roughly one in every ten discharges carried a renal failure diagnosis (Figure 13). Figure 14 summarizes the composition of the cohort, the split on the dependent variable, and the extent of missing data on each predictor.

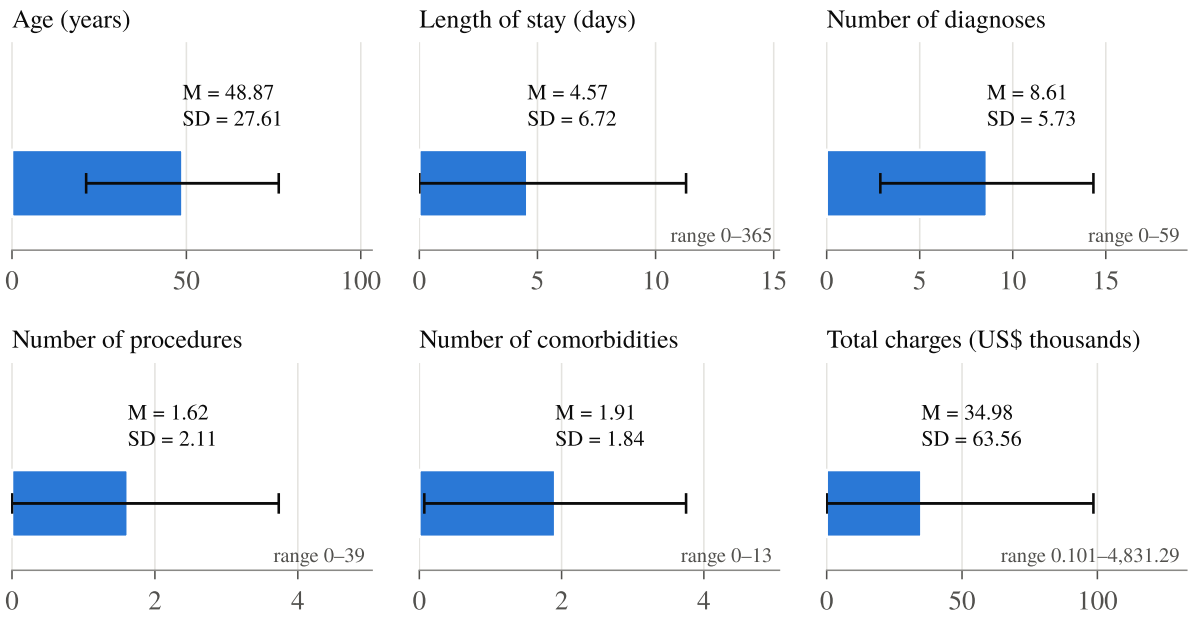


Figure 12

Mean and Standard Deviation of the Continuous Predictors

Note. Bars show the mean; whiskers show ±1 *SD* (truncated at zero). Values from Tables 5 and 7 and the reported mean age (*M* = 48.87, *SD* = 27.61). Panels use separate scales because the variables are measured in different units.

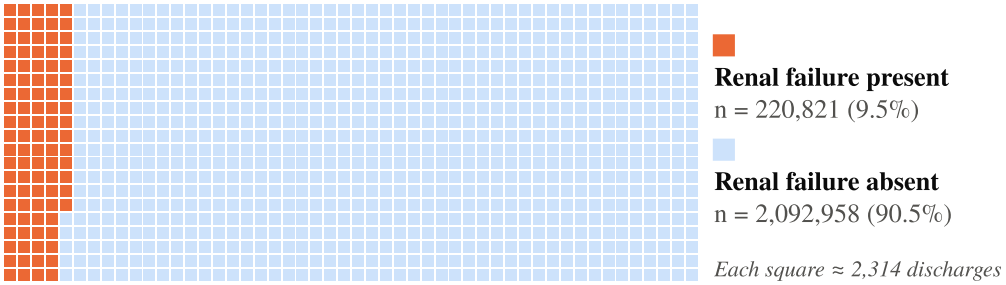


Figure 13

Prevalence of Renal Failure Among Inpatient Discharges

Note. Each of the 1,000 squares represents 0.1% of the sample. Data from Table 6.

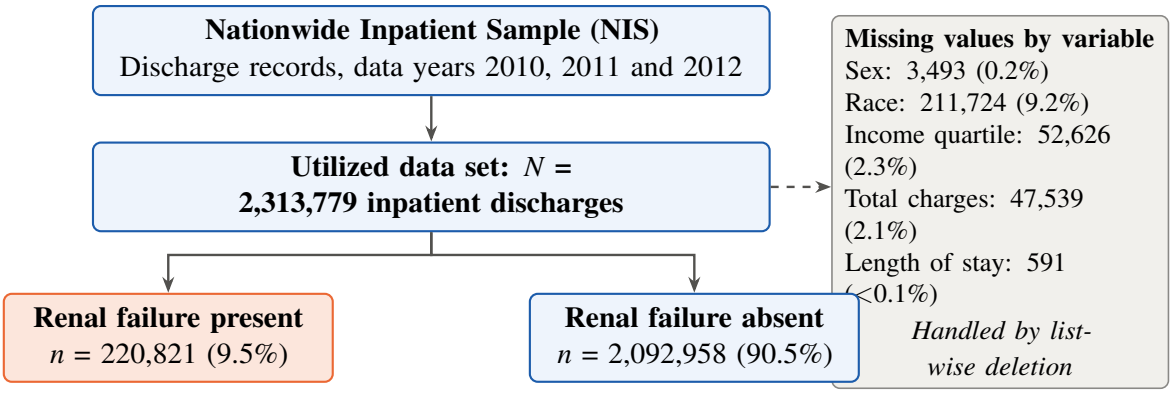


Figure 14

Composition of the Study Cohort and Distribution of the Dependent Variable

Note. Counts are taken from Tables 2–6; missing counts for total charges and length of stay are derived as 2,313,779 minus the valid *N* reported in Table 5.

The remaining 27 comorbidities were measured as a dichotomous variable indicating the presence (1) or absence (0). These variables were summed together and resulted in an overall measure of the number of comorbidities. The number of comorbidities ranged from 0 to 13 (*M* = 1.91, *SD* = 1.84). See Table 7.

Table 7

Number of Comorbidities

	Minimum	Maximum	*M*	*SD*
Number of comorbidities	0.00	13.00	1.91	1.84

Assumptions of Logistic Regression

Binary logistic regression addressed the study's research questions and hypotheses. Before the analysis, there were some assumptions. The assumptions for logistic regression included linearity in the logit for continuous variables, absence of multicollinearity, and lack of strongly influential outliers (Field, 2018). The continuous variables' linearity concerning the dependent variable's logit was assessed via the Box-Tidwell procedure. Based on this assessment, all continuous independent variables were found to be linearly related to the logit of the dependent

variable. There were no variance inflation factors (VIFs) beyond 10, thus no multicollinearity. There were no standardized residuals beyond three standard deviations, thus no outliers.

Results

RQ1: Is there a relationship between renal failure and length of stay?

H_0: There is no significant relationship between renal failure and length of stay.
H_1: There is a significant relationship between renal failure and length of stay.

The logistic regression model was statistically significant, $\chi^2(1) = 11383.027$, $p < .001$. Length of stay was a significant predictor of renal failure ($B = 0.028$, OR = 1.028, $p < .001$). A one-day increase in length of stay increased the odds of renal failure by 1.028 times. Table 8 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and length of stay.

Table 8
Results of Logistic Regression (LOS Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
LOS	.028	.000	11,338.115	1	.000	1.028	1.028	1.029
Constant	−2.390	.003	806,282.546	1	.000	.092		

Note. Model $\chi^2(1) = 11{,}383.027$, $p < .001$. OR = odds ratio; CI = confidence interval. LOS = length of stay (days).

RQ2: Is there a relationship between renal failure and total charges?

H_0: There is no significant relationship between renal failure and total charges.
H_1: There is a significant relationship between renal failure and total charges.

The logistic regression model was statistically significant, $\chi^2(1) = 11016.100$, $p < .001$. Total charge was a significant predictor of renal failure ($B = 0.000003$, OR = 1.000003, $p < .001$). A one-unit increase in total charged increased the odds of renal failure by 1.000003 times. Table 9 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and total charge.

Table 9

Results of Logistic Regression (Total Charges Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
TOTCHG	.000	.000	10,939.729	1	.000	1.000	1.000	1.000
Constant	−2.372	.003	846,375.720	1	.000	.093		

Note. Model $\chi^2(1)$ = 11,016.100, $p < .001$. OR = odds ratio; CI = confidence interval. TOTCHG = total charges in dollars; unrounded B = 0.000003 (OR = 1.000003 per dollar, equivalent to approximately 1.03 per $10,000).

RQ3: Is there a relationship between renal failure and age?

H_0: There is no significant relationship between renal failure and age.
H_1: There is a significant relationship between renal failure and age.

The logistic regression model was statistically significant, $\chi^2(1)$ = 181703.603, $p < .001$. Age significantly predicted renal failure (B = 0.044, OR = 1.045, $p < .001$). A one-unit increase in age increased the odds of renal failure by 1.045 times. Table 10 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and age.

Table 10

Results of Logistic Regression (Age Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
AGE	.044	.000	125,392.064	1	.000	1.045	1.045	1.046
Constant	−4.921	.009	303,010.744	1	.000	.007		

Note. Model $\chi^2(1)$ = 181,703.603, $p < .001$. OR = odds ratio; CI = confidence interval.

RQ4: Is there a relationship between renal failure and gender?

H_0: There is no significant relationship between renal failure and gender.
H_1: There is a significant relationship between renal failure and gender.

The logistic regression model was statistically significant, $\chi^2(1)$ = 11,132.04, $p < .001$. Gender (Female) was a significant predictor of renal failure (B = −0.477, OR = 0.621, $p <$

.001). Being female reduced the likelihood of renal failure by 0.621 times. In other words, males are 1/0.621 = 1.61 times as likely as females to have renal failure. Table 11 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and gender.

Table 11

Results of Logistic Regression (Gender Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
Female	−.477	.005[a]	11,122.002[a]	1	.000	.621	.615[a]	.626[a]
Constant	−2.013	.003[a]	410,923.686[a]	1	.000	.134		

Note. Model $\chi^2(1)$ = 11,132.04[a], $p < .001$. OR = odds ratio; CI = confidence interval. Reference category: male. [a]Corrected value, reconstructed from the reported coefficients and the sex distribution in Table 2 by maximum likelihood; the same procedure reproduces the published statistics of the race model (Table 12) to within rounding error.

RQ5: Is there a relationship between renal failure and race?

H_0: There is no significant relationship between renal failure and race.
H_1: There is a significant relationship between renal failure and race.

The logistic regression model was statistically significant, $\chi^2(5) = 8515.693$, $p < .001$. Race was a significant predictor of renal failure. Specifically, compared to White, Blacks were 1.539 times as likely to have renal failure ($B = 0.431$, OR = 1.539, $p < .001$). Additionally, compared to White, Hispanics ($B = -.310$, OR = 0.733, $p < .001$); Asian ($B = -.083$, OR = 0.920, $p < .001$); Native American ($B = -.062$, OR = 0.940, $p = .035$); and some other race ($B = -.328$, OR = 0.720, $p < .001$) were less likely to have renal failure. Table 12 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and race.

Table 12

Results of Logistic Regression (Race Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
White (reference)			8,737.625	5	.000			
Black	.431	.006	5,347.199	1	.000	1.539	1.521	1.557
Hispanic	−.310	.008	1,402.768	1	.000	.733	.721	.745
Asian	−.083	.016	28.484	1	.000	.920	.893	.949
Native American	−.062	.029	4.456	1	.035	.940	.888	.996
Other	−.328	.015	479.974	1	.000	.720	.699	.742
Constant	−2.270	.003	605,950.398	1	.000	.103		

Note. Model $\chi^2(5)$ = 8,515.693, $p < .001$. OR = odds ratio; CI = confidence interval.

RQ6: Is there a relationship between renal failure and socioeconomic status (SES)?

H_0: There is no significant relationship between renal failure and socioeconomic status (SES).

H_1: There is a significant relationship between renal failure and socioeconomic status (SES).

The logistic regression model was statistically significant, $\chi^2(1) = 756.98$, $p < .001$. SES was a significant predictor of renal failure, and increasing SES decreased the likelihood of renal failure ($B = -.056$, OR = 0.946, $p < .001$). Table 13 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and SES.

Table 13

Results of Logistic Regression (SES Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
SES	−.056	.002	752.588	1	.000	.946	.942	.949
Constant	−2.115	.005	164,321.338	1	.000	.121		

Note. Model $\chi^2(1) = 756.98^{a}$, $p < .001$. OR = odds ratio; CI = confidence interval. [a]Corrected value, reconstructed from the reported coefficients and the quartile distribution in Table 4 by maximum likelihood; all other values in the table are as estimated. SES = median household income national quartile for patient ZIP Code, entered as an ordinal score (1 = poorest to 4 = wealthiest).

RQ7: Is there a relationship between renal failure and the number of procedures?

H_0: There is no significant relationship between renal failure and the number of procedures.
H_1: There is a significant relationship between renal failure and the number of procedures.

The logistic regression model was statistically significant, $\chi^2(1) = 6120.885$, $p < .001$. The number of procedures was a significant predictor of renal failure. The increasing number of procedures increases the likelihood of renal failure ($B = 0.074$, OR = 1.077, $p < .001$). Table 14 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and the number of procedures.

Table 14
Results of Logistic Regression (Number of Procedures Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
NPR	.074	.001	6,721.985	1	.000	1.077	1.075	1.079
Constant	−2.380	.003	704,974.741	1	.000	.093		

Note. Model $\chi^2(1) = 6{,}120.885$, $p < .001$. OR = odds ratio; CI = confidence interval. NPR = number of ICD-9-CM procedures.

RQ8: Is there a relationship between renal failure and the number of diagnoses?

H_0: There is no significant relationship between renal failure and the number of diagnoses.
H_1: There is a significant relationship between renal failure and the number of diagnoses.

The logistic regression model was statistically significant, $\chi^2(1) = 269839.558$, $p < .001$. The number of diagnoses was a substantial predictor of renal failure. The increasing number of diagnoses increases the likelihood of renal failure ($B = 0.190$, OR = 1.209, $p < .001$). Table 15 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and the number of diagnoses.

RQ9: Is there a relationship between renal failure and comorbidities?

H_0: There is no significant relationship between renal failure and comorbidities.
H_1: There is a significant relationship between renal failure and comorbidities.

Table 15

Results of Logistic Regression (Number of Diagnoses Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
NDX	.190	.000	228,123.380	1	.000	1.209	1.208	1.210
Constant	−4.370	.006	560,368.427	1	.000	.013		

Note. Model $\chi^2(1)$ = 269,839.558, $p < .001$. OR = odds ratio; CI = confidence interval. NDX = number of ICD-9-CM diagnoses.

The logistic regression model was statistically significant, $\chi^2(1) = 210680.981$, $p < .001$. The number of comorbidities was a significant predictor of renal failure. The increasing number of comorbidities increases the likelihood of renal failure ($B = 0.523$, OR = 1.687, $p < .001$). Table 16 provides this information. Thus, the null hypothesis is rejected and concludes that there is a significant relationship between renal failure and the number of comorbidities.

Table 16

Results of Logistic Regression (Number of Comorbidities Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
Number of comorbidities	.523	.001	187,824.244	1	.000	1.687	1.683	1.691
Constant	−3.634	.005	610,905.820	1	.000	.026		

Note. Model $\chi^2(1)$ = 210,680.981, $p < .001$. OR = odds ratio; CI = confidence interval.

Visual Summary of the Individual Predictor Models (RQ1–RQ9)

Figure 15 brings the nine unadjusted models (Tables 8–16) together on a common odds-ratio scale. Because the odds ratios of continuous predictors are expressed per unit (one day, one year, one diagnosis), their magnitudes depend on the unit chosen; total charges is therefore shown per $10,000 rather than per dollar. Among the categorical predictors, Black race carried the largest increase in odds relative to White patients, and female sex the largest decrease.

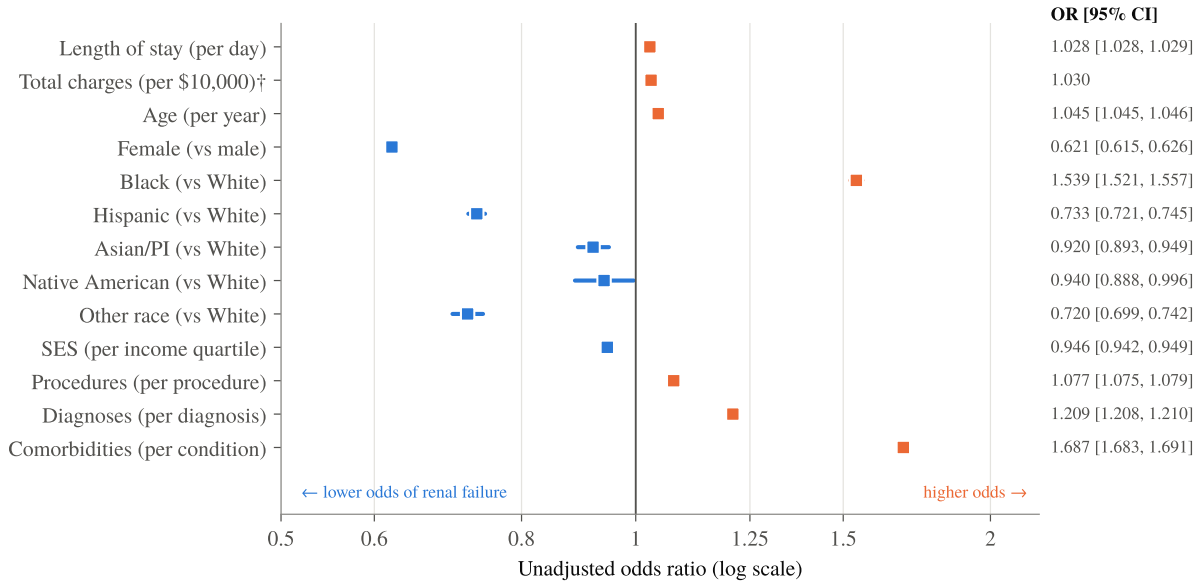


Figure 15

Unadjusted Odds Ratios for Renal Failure From the Nine Individual Logistic Regression Models

Note. Squares show odds ratios and horizontal lines their 95% confidence intervals (Tables 8–16); most intervals are narrower than the marker because of the very large sample. †Derived as $e^{0.000003 \times 10{,}000}$; no interval was reported. The interval for sex is the corrected interval in Table 11.

To translate the coefficients into a more intuitive metric, the predicted probability of renal failure was computed from each reported model as $\hat{p} = 1/\{1 + \exp[-(\beta_0 + \beta_1 x)]\}$, using the constant ($\beta_0$) and slope ($\beta_1$) in Tables 8–16. Figure 16 shows these curves over the clinically plausible range of each continuous predictor. The steepest gradients are for the number of comorbidities and the number of diagnoses: the predicted probability rises from about 3% with no comorbidities to above 50% with seven or more, and reaches 50% at about 23 recorded diagnoses. Figure 17 shows the corresponding predicted probabilities for the categorical predictors.

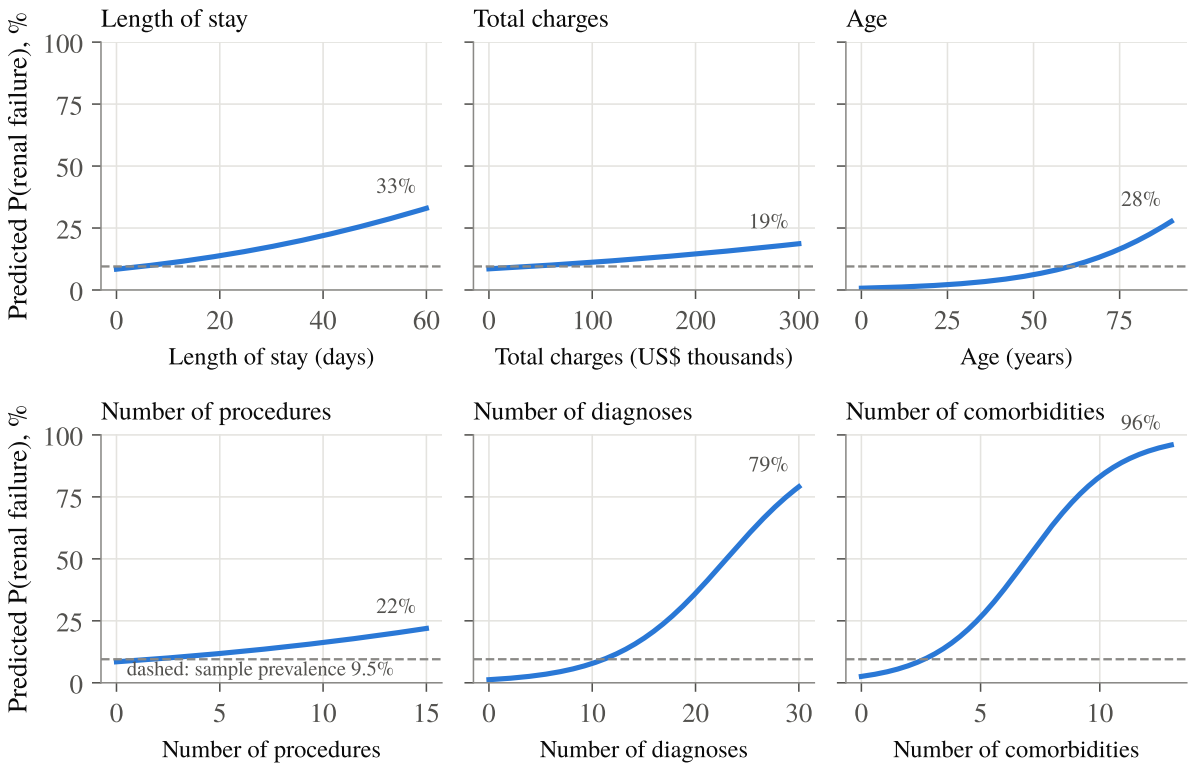


Figure 16

Predicted Probability of Renal Failure Across the Range of Each Continuous Predictor (Unadjusted Models)

Note. Curves derived from the constants and coefficients reported in Tables 8, 9, 10, 14, 15, and 16. The dashed line marks the sample prevalence (9.5%). The labels give the predicted probability at the right-hand end of each axis. Each curve reflects one predictor at a time and is not adjusted for the other variables.

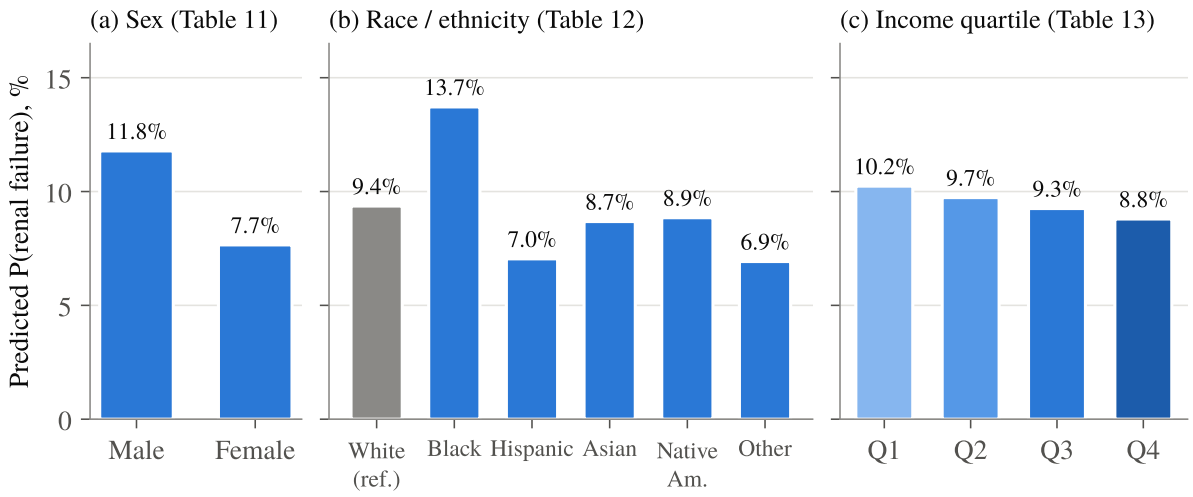


Figure 17

Predicted Probability of Renal Failure by Sex, Race/Ethnicity, and Income Quartile (Unadjusted Models)

Note. Derived from Tables 11–13 as $1/\{1+\exp[-(\beta_0+\beta_1)]\}$. In panel (c), SES was modeled as a linear score, so the quartile differences are equally spaced on the logit scale.

RQ10: Do length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively predict renal failure?

H_0: Length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively do not significantly predict renal failure.

H_1: Length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively significantly predict renal failure.

The logistic regression model was statistically significant, $\chi^2(15) = 310203.539$, $p < .001$. The model explained 30.5% (Nagelkerke R^2) of the variance in renal failure and correctly classified 90.3% of cases. The regression coefficients appear in Table 17. Thus, this null hypothesis is rejected and concludes that length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and co-morbidities collectively significantly predict renal failure.

The adjusted odds ratios in Table 17 are plotted in Figure 18. Holding the other variables constant, Black race (AOR = 2.609) was the strongest single risk marker, followed by the other minority groups (AOR = 1.318–1.753), each additional comorbidity (AOR = 1.149) and each

Table 17

Results of Logistic Regression (Length of Stay, Total Charges, Age, Gender, Race, Socioeconomic Status (SES), Number of Procedures, Number of Diagnoses, and Comorbidities Predicting Renal Failure)

	B	*SE*	Wald	*df*	*p*	OR	95% CI for OR	
							Lower	Upper
LOS	−.010	.001	365.297	1	.000	.990	.989	.991
TOTCHG	.000	.000	28.817	1	.000	1.000	1.000	1.000
AGE	.033	.000	40,747.592	1	.000	1.034	1.034	1.034
FEMALE	−.485	.005	8,143.837	1	.000	.616	.609	.622
White (reference)			18,492.530	5	.000			
Black	.959	.007	17,653.485	1	.000	2.609	2.572	2.646
Hispanic	.488	.010	2,447.021	1	.000	1.629	1.598	1.661
Asian	.561	.019	880.316	1	.000	1.753	1.689	1.819
Native American	.407	.034	141.701	1	.000	1.502	1.404	1.606
Other	.276	.017	257.374	1	.000	1.318	1.274	1.363
SES 1st quartile (reference)			330.223	3	.000			
SES 2nd quartile	−.021	.007	8.296	1	.004	.979	.966	.993
SES 3rd quartile	−.065	.007	77.243	1	.000	.937	.924	.951
SES 4th quartile	−.136	.008	294.554	1	.000	.873	.860	.887
NPR	.002	.001	1.868	1	.172	1.002	.999	1.004
NDX	.133	.001	42,066.414	1	.000	1.142	1.141	1.144
Number of comorbidities	.139	.002	5,087.340	1	.000	1.149	1.145	1.153
Constant	−6.080	.014	197,835.081	1	.000	.002		

Note. Model $\chi^2(15) = 310{,}203.539$, $p < .001$; Nagelkerke $R^2 = .305$; 90.3% of cases correctly classified; AUC = .856. OR = odds ratio; CI = confidence interval. LOS = length of stay; TOTCHG = total charges; NPR = number of procedures; NDX = number of diagnoses.

additional diagnosis (AOR = 1.142). Female sex (AOR = 0.616) and higher income quartiles (AOR = 0.873 for the wealthiest quartile) were associated with lower odds. The number of procedures was the only predictor that did not reach significance in the adjusted model (p = .172).

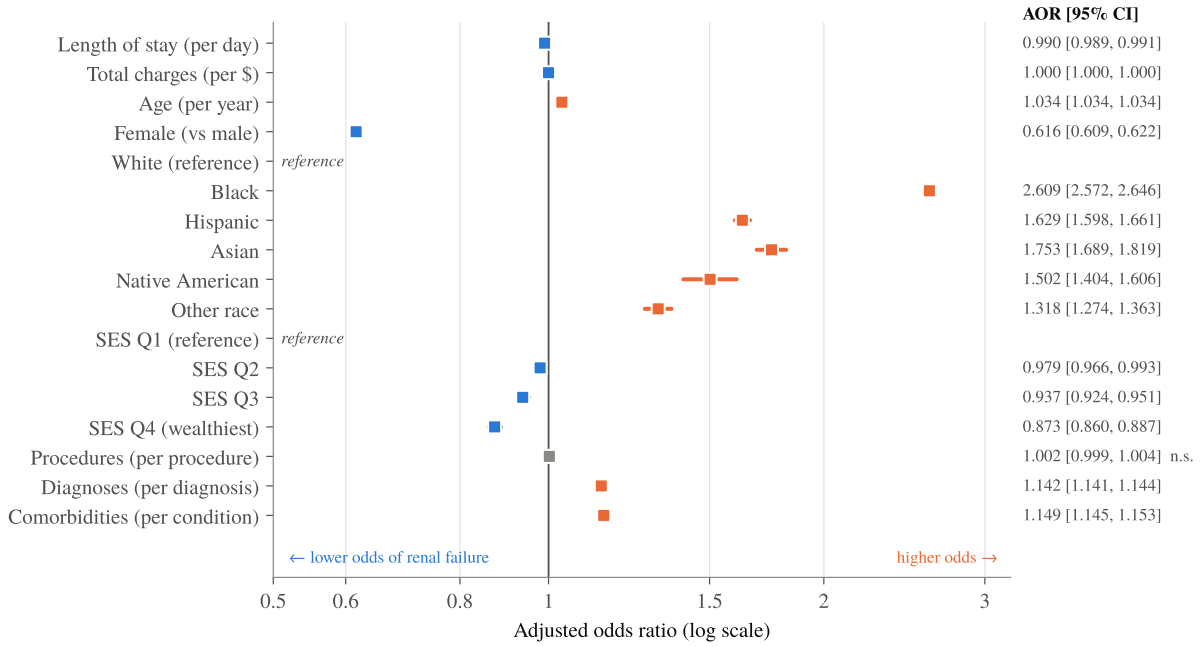


Figure 18

Adjusted Odds Ratios for Renal Failure From the Multivariable Logistic Regression Model (RQ10)

Note. Data from Table 17. Orange = significantly higher odds; blue = significantly lower odds; grey = not significant (n.s., $p > .05$). AOR = adjusted odds ratio.

Figure 19 ranks the predictors by their Wald statistic in the multivariable model, a measure of how strongly each one contributes once the others are taken into account. The number of diagnoses and age contributed the most, followed by race, sex, and the number of comorbidities.

The Receiver Operating Characteristic (ROC) curve was produced to assess further the model's collective predictability (Figure 20). The area under the ROC curve was .856, which is an excellent level of discrimination, according to Hosmer et al. (2013).

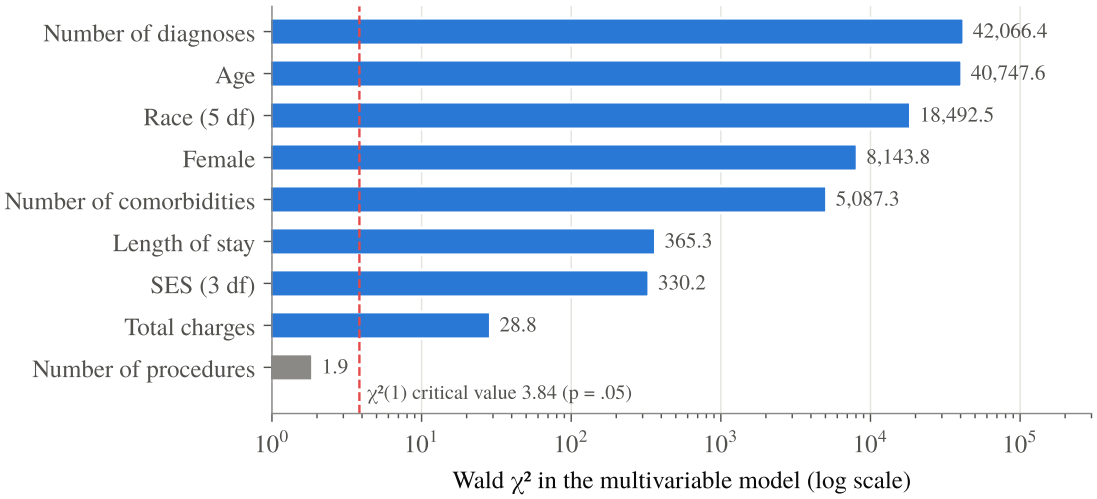


Figure 19

Relative Contribution of Each Predictor to the Multivariable Model (Wald Chi-Square)

Note. Wald statistics from Table 17, plotted on a logarithmic scale. Race and SES are shown as their omnibus (multi-degree-of-freedom) tests. The dashed line marks the critical value for a 1-*df* test at α = .05.

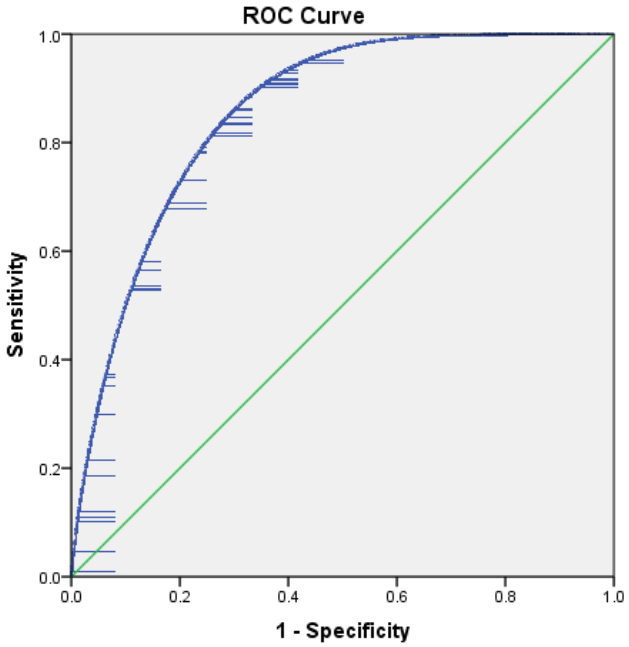


Figure 20

ROC Curve (Length of Stay, Total Charges, Age, Gender, Race, Socioeconomic Status (SES), Number of Procedures, Number of Diagnoses, and Comorbidities Predicting Renal Failure)

Note. The diagonal reference line represents chance discrimination (AUC = .50). Area under the curve = .856.

Figure 21 places the model's performance statistics in context. The AUC of .856 falls within the excellent band of Hosmer et al. (2013). Because renal failure is present in only 9.5% of cases, a model that predicted "absent" for every patient would already classify 90.5% correctly. The overall classification rate (90.3%) should therefore be read alongside the AUC, which is not affected by the outcome's prevalence.

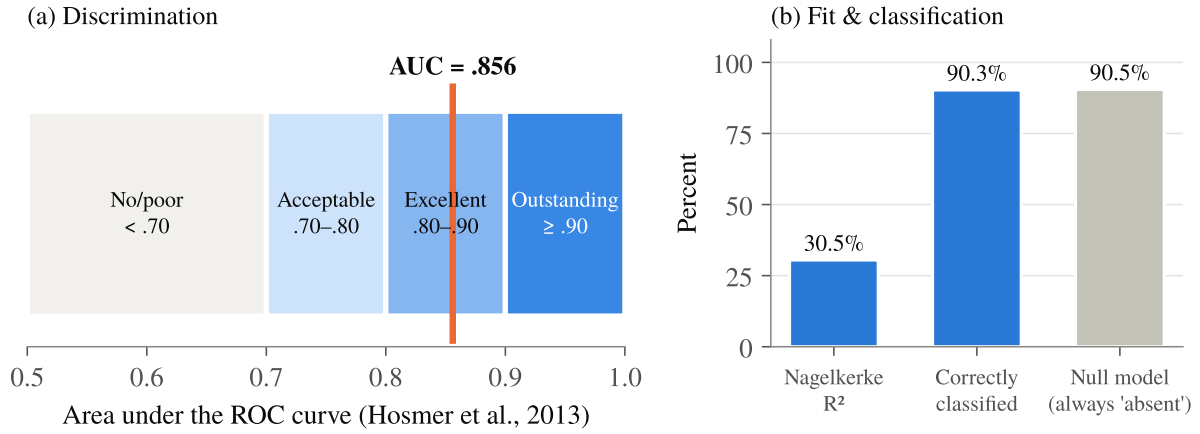


Figure 21

Discrimination, Explained Variance, and Classification Accuracy of the Multivariable Model

Note. AUC interpretation bands follow Hosmer et al. (2013). The null-model rate (90.5%) is the proportion of cases without renal failure (Table 6); it is shown for comparison.

Comparison of Unadjusted and Adjusted Estimates

Comparing the single-predictor models (RQ1–RQ9) with the multivariable model (RQ10) shows how much each association depends on the other variables (Figure 22). Three patterns stand out. First, the associations for age, sex, the number of diagnoses and SES kept their direction and remained similar in size. Second, the odds ratio for comorbidities fell from 1.687 to 1.149, and the odds ratio for the number of procedures fell from 1.077 to a non-significant 1.002. This indicates that much of their unadjusted association is shared with the number of diagnoses and with age. Third, two associations reversed direction. Length of stay moved from a positive (OR = 1.028) to a slightly negative (AOR = 0.990) association. The Hispanic, Asian, Native American and Other race groups, which had lower unadjusted odds than White patients, had higher odds once age and the other variables were held constant. A plausible explanation for

the race reversal is confounding. For example, if these groups are younger on average in the sample, adjusting for age would raise their estimated odds. The reported output does not allow this to be verified directly. Readers should therefore treat the adjusted estimates in Table 17 as the primary basis for inference about each predictor.

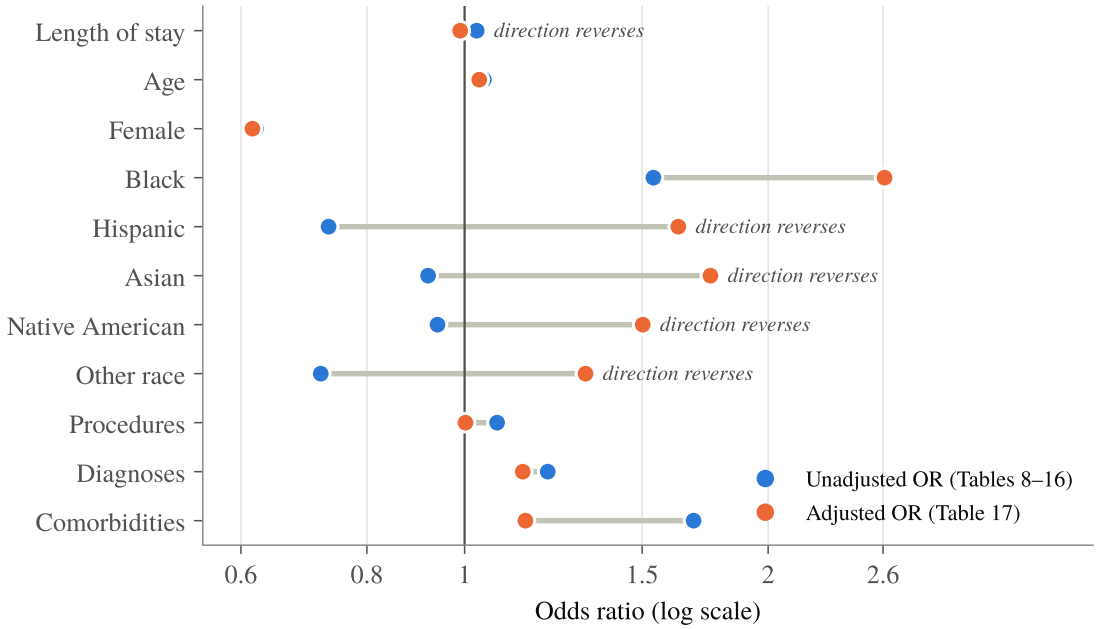


Figure 22

Change in Odds Ratios From the Unadjusted (RQ1–RQ9) to the Adjusted (RQ10) Models

Note. Blue points: unadjusted ORs (Tables 8–16); orange points: adjusted ORs (Table 17). Total charges (OR $\approx$ 1.000 per dollar in both models) and SES (coded differently in the two models) are omitted.

Summary

The purpose of this study was to investigate the predictors of renal failure based on length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and co-morbidities in a cohort of inpatients in the United States. The ten research questions (RQ1–RQ10) were addressed; the decision for each is summarized in Table 18.

In the individual (unadjusted) models, the likelihood of renal failure increased significantly with increasing length of stay, total charges, age, number of procedures, number of diagnoses, and number of comorbidities, as well as for male patients and for Black patients

Table 18

Summary of Hypothesis Tests for Research Questions 1–10

RQ	Predictor	Model test	Unadjusted OR	Adjusted OR	Decision
RQ1	Length of stay	$\chi^2(1) = 11{,}383.03$	1.028 per day	0.990 per day	Reject H_0
RQ2	Total charges	$\chi^2(1) = 11{,}016.10$	1.000003 per $	1.000 per $	Reject H_0
RQ3	Age	$\chi^2(1) = 181{,}703.60$	1.045 per year	1.034 per year	Reject H_0
RQ4	Gender (female)	$\chi^2(1) = 11{,}132.04$[a]	0.621	0.616	Reject H_0
RQ5	Race (Black vs. White)	$\chi^2(5) = 8{,}515.69$	1.539	2.609	Reject H_0
RQ6	SES (income quartile)	$\chi^2(1) = 756.98$[a]	0.946 per quartile	0.873 (Q4 vs. Q1)	Reject H_0
RQ7	Number of procedures	$\chi^2(1) = 6{,}120.89$	1.077	1.002 (n.s.)	Reject H_0[b]
RQ8	Number of diagnoses	$\chi^2(1) = 269{,}839.56$	1.209	1.142	Reject H_0
RQ9	Number of comorbidities	$\chi^2(1) = 210{,}680.98$	1.687	1.149	Reject H_0
RQ10	All predictors jointly	$\chi^2(15) = 310{,}203.54$	—	AUC = .856	Reject H_0

Note. All model tests $p < .001$. Unadjusted ORs from Tables 8–16; adjusted ORs from Table 17. [a]Value reconstructed by maximum likelihood from the reported coefficients and group sizes (Tables 2 and 4). [b]Significant in the unadjusted model (RQ7) but not in the adjusted model ($p = .172$).

(compared to White), and decreased significantly with increasing SES. When all predictors were entered together, they collectively predicted renal failure with excellent discrimination (AUC = .856). In this adjusted model, older age, male sex, every minority racial group (compared to White), lower SES, and greater numbers of diagnoses and comorbidities remained independent predictors of renal failure; length of stay showed a small inverse association (AOR = 0.990), and the number of procedures was no longer statistically significant.

What follows in Chapter 5 is a discussion of how the results of this study are interpreted in the context of the theoretical framework. Any limitations of the results of the study will be provided. Additionally, recommendations for future research will be discussed.

CHAPTER V: DISCUSSION, RECOMMENDATIONS, AND CONCLUSION

The purpose of this study was to examine the relationship between renal failure and length of stay, total charges, and their association with age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities in a cohort of inpatients in the United States. This study's research methodology and design was a quantitative correlational design study. A quantitative research methodology is based on an objectivity paradigm, uses numerical data to enable statistical analyses, and helps minimize biases (Bowers, 2017). This study's data collection instrument was archival data of the selected inpatients in the United States.

After data analysis, the results revealed the following. In the individual (unadjusted) models, the likelihood of renal failure increased significantly with increasing length of stay, total charges, age, number of procedures, number of diagnoses, and number of comorbidities, as well as for male patients and for Black patients (compared to White), and decreased significantly with increasing SES. When all predictors were entered together, they collectively predicted renal failure with excellent discrimination (AUC = .856). In this adjusted model, older age, male sex, every minority racial group (compared to White), lower SES, and greater numbers of diagnoses and comorbidities remained independent predictors of renal failure; length of stay showed a small inverse association (AOR = 0.990), and the number of procedures was no longer statistically significant. Chapter 5 presents a discussion of how the results of this study are interpreted in the context of the theoretical framework. Any limitations of the results of the study were provided. Additionally, recommendations for future research were discussed. The chapter concludes with the study conclusion.

Discussion and Interpretation of Findings

The discussion and interpretation of findings were based on the research questions described below.

Table 19 summarizes how each finding relates to the literature discussed below, and Figure 23 summarizes the direction of the adjusted associations.

Table 19

Alignment of the Study Findings With the Prior Literature Discussed in Chapter V

RQ	Predictor	Direction found	Agreement	Key sources cited in the discussion
RQ1	Length of stay	Higher odds (unadjusted)	Consistent	Chang et al. (2022); Soldati et al. (2022); Hurst et al. (2022)
RQ2	Total charges	Higher odds	Consistent	Lv & Zhang (2019); Morton & Shah (2020); Freeman et al. (2019)
RQ3	Age	Higher odds	Consistent	Hurst et al. (2022); Wilkinson et al. (2022); Otobe et al. (2022); Mei et al. (2021)
RQ4	Gender	Lower odds in women	Inconsistent	Tong et al. (2022); Garcia et al. (2022); Vinson (2022); Smothers et al. (2022)
RQ5	Race	Higher odds in Black patients	Consistent	Eneanya et al. (2022); Crews et al. (2022); Boulware & Mohottige (2021)
RQ6	SES	Lower odds with higher SES	Consistent	Morton (2022); Novick et al. (2022); Saunders et al. (2020)
RQ7	Number of procedures	Higher odds (unadjusted)	Consistent	Schrauben et al. (2021); El Shamy & Cavanaugh (2020)
RQ8	Number of diagnoses	Higher odds	Consistent	Shah et al. (2021); Chu et al. (2020)
RQ9	Comorbidities	Higher odds	Consistent	Jacob et al. (2019); Sullivan et al. (2020); MacRae et al. (2020)
RQ10	Joint model	Significant joint prediction	Consistent	Tong et al. (2022); Soldati et al. (2022); Morton & Shah (2020); Sullivan et al. (2020)

Note. Agreement is classified as described in the discussion of each research question. For RQ4, the finding that men had higher odds contradicts the access-based literature on women, but it agrees with studies that report greater susceptibility in men (Duan et al., 2019; Cañadas-Garre et al., 2019; Valdivielso et al., 2019) reviewed in Chapter II.

RQ1: Is there a relationship between renal failure and length of stay?

Data analysis established a significant relationship between the length of stay and renal failure. The length of stay was a significant predictor of renal failure, such that an increase in length of

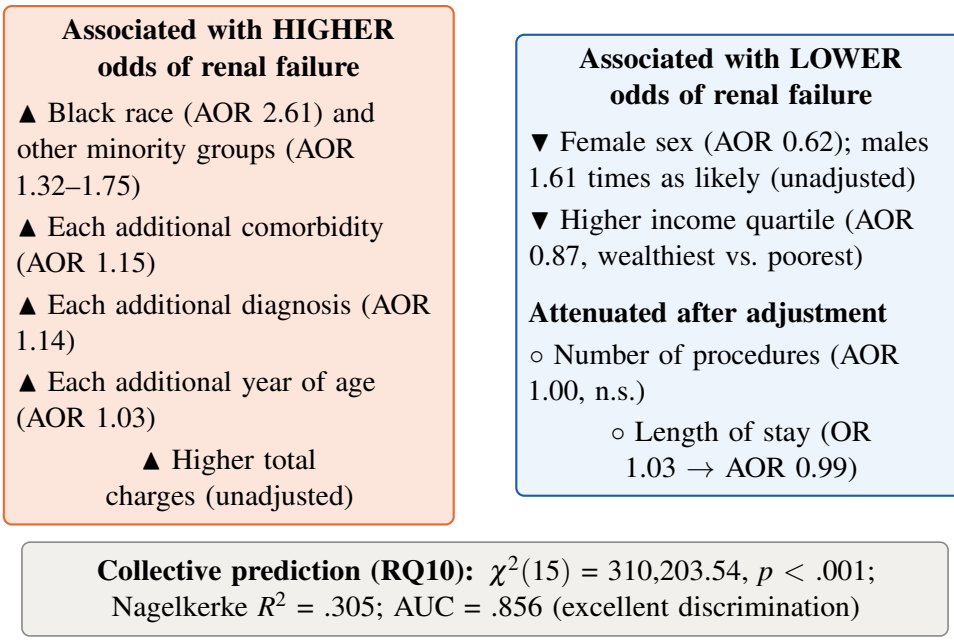


Figure 23
Key Findings at a Glance

Note. AOR = adjusted odds ratio from Table 17; unadjusted values from Tables 8–16.

stay increased the odds of renal failure. The results imply that patients with kidney disease who stay longer in hospitals are more likely to experience renal failure because the length of stay may determine the severity of kidney disease among the patients. However, previous literature established that patients with chronic kidney disease suffered from frailty, evidenced by old age, comorbidity, and limited social care (Chang et al., 2022).

Soldati et al. (2022) also established higher frailty index in older people was typified by increased hospitalization and hospital stay due to chronic kidney disease. Frailty affects the nervous system and interferes with the functioning of vital organs like the kidneys, thus increasing susceptibility to chronic diseases such as renal failure. Although current study findings demonstrated the impact of length of stay on renal failure, Hurst et al. (2022) found that frailty and renal failure in older populations were associated with frequent hospitalization and increased length of stay. The findings have contributed to previous literature by establishing that increased length of stay significantly increases renal failure among patients with kidney disease.

RQ2: Is there a relationship between renal failure and total charge?

The logistic regression model was statistically significant, and the findings demonstrated that the total charge was a significant predictor of renal failure. The results indicated that an increase in the total charge contributes to a significant increase in renal failure among patients. As a result, the null hypothesis was rejected, and it concluded that there is a significant relationship between renal failure and total charges. The implication is that the total charged may be a significant predictor of renal failure because the more complex the kidney disease is, the total cost charged for treatment. Increased renal failure is associated with high chronic kidney disease, which requires a high cost of treatment among patients.

Consistent with current study results, previous research established that the high total costs of treatment and associated chronic diseases such as hypertension, aging, and diabetes mellitus increased financial burden are associated with renal failure among patients (Lv & Zhang, 2019). Due to high healthcare costs, many middle-income individuals cannot afford the cost of chronic kidney disease treatment and the treatment of other chronic illnesses, such as hypertension and diabetes, resulting in renal failure (Morton & Shah, 2020). Although current study findings demonstrated total charges as the key predictor of renal failure, previous literature highlighted that continued increases in treatment make people feel economically stretched and incapable of affording medication costs, resulting in increased renal failure (Freeman et al., 2019). The findings have added to previous literature by revealing that the total cost charged for kidney disease treatment could significantly predict renal failure among patients.

RQ3: Is there a relationship between renal failure and age?

Age was a significant predictor of renal failure, and renal failure odds increased with a corresponding increase in the patient's age with kidney disease problems. As a result, the null hypothesis was rejected, and it concluded that there was a significant relationship between renal failure and age. The findings imply that age predicted the possibility of developing renal failure among patients with chronic kidney disease. The implication is that renal failure

increases with the patient's age or victim of kidney disease. The findings are consistent with the previous empirical literature. Hurst et al. (2022) found that frailty in older populations was associated with frequent hospitalization of older people with chronic kidney disease and other non-communicable diseases resulting in renal failure.

Current study results also confirm Wilkinson et al. (2022), who reported that frailty was prevalent in older people suffering from chronic kidney disease resulting in renal failure. Older populations are more likely to suffer from renal failure than younger populations (Wilkinson et al., 2022). Wilkinson et al. (2022) findings confirm current research results indicating that age significantly predicts renal failure. Similar to current findings, Otobe et al. (2022) established that frailty in older people was likely to increase the prevalence of chronic kidney disease cases due to reduced organ function, resulting in renal failure among such patients. Older patients suffering from chronic kidney disease had a predictor of negative outcomes of chronic kidney disease, such as mortality, increased hospitalization, and renal failure (Mei et al., 2021). Research findings have added to the previous literature by revealing that age significantly predicts renal failure among patients suffering from chronic kidney diseases.

RQ4: Is there a relationship between renal failure and gender?

Gender (Female) was a significant predictor of renal failure, and the results revealed that female patients had a reduced likelihood of renal failure. In other words, males are more likely than females to have renal failure. The implication is that gender significantly impacts renal failure, and males are likelier to suffer renal failure than female patients with reduced renal failure. Consistent with current research findings, previous literature highlighted that chronic kidney disease was more prevalent in women than men as there was limited access to kidney healthcare by women as a result of stereotyping by clinicians and economic disadvantage (Tong et al., 2022). Although current findings did not indicate why female patients experience reduced renal failure, the result contradicts prior literature, which revealed that women received limited kidney healthcare services compared to men because of a lack of medical insurance coverage, gender stereotyping in health centers, and cultural and social norms (Garcia et al., 2022).

Research findings are inconsistent with previous literature. While current findings revealed female patients had reduced renal failure compared to men, past research indicated that women received limited kidney healthcare services compared to men because of a lack of medical insurance coverage and cultural and social norms, thereby resulting in renal failure among women compared to men (Vinson, 2022). In contrast to current research findings, Smothers et al. (2022) established that women were less likely to be referred to a donor than men, increasing the prevalence of chronic kidney diseases and subsequent renal failure among women. Women are more likely than men to suffer from gender inequities in access to kidney healthcare, contributing to the higher prevalence of chronic kidney disease in women (Smothers et al., 2022). The findings have contributed to past research by revealing that gender is a significant predictor of renal failure and that female patients are less likely to suffer from renal failure than men.

RQ5: Is there a relationship between renal failure and race?

The findings revealed that race was a significant predictor of renal failure. Specifically, compared to Whites, Blacks were more likely to have renal failure. Additionally, compared to White, Hispanics, Asians, Native Americans, and some other races were less likely to have renal failure. Results imply that race has a significant impact on renal failure. The implication is that patients of different races experience renal failure conditions differently and that minority races are more affected by renal failure compared to the majority, the whites. Current results concur with previous research findings, highlighting that black individuals were more likely to suffer from chronic kidney disease and renal failure than whites propagated by racism, leading to black individuals having limited access to affordable and conclusive healthcare services (Eneanya et al., 2022). The authors confirm current research findings that race significantly predicts renal failure (Eneanya et al., 2022).

The prevalence of chronic kidney disease is influenced by racism, such that structural racism has increased chronic kidney disease among black people compared to other races, perpetuated by a lack of medical services and comprehensive treatment for renal diseases

(Crews et al., 2022). Whereas current research opined that whites were less likely to suffer from renal failure and kidney chronic kidney diseases, Boulware and Mohottige (2021) studied the racial and social inequities affecting kidney care and reported that ethnic minority groups such as blacks, Hispanics, Asians, and Native Americans were less likely to receive timely dialysis referrals than non-minorities, thus worsening the prevalence of chronic kidney disease resulting renal failure. The results have contributed to the previous literature by establishing that race is a significant predictor of renal failure among patients suffering from chronic kidney diseases.

RQ6: Is there a relationship between renal failure and socioeconomic status (SES)?

Socioeconomic status was found to be a significant predictor of renal failure such that increasing socioeconomic status results in decreased likelihood of renal failure among patients suffering from kidney disease. The results imply that patients' likelihood of suffering from renal failure was determined by their socioeconomic statuses, such as income level and education level. The findings are consistent with past literature. Morton (2022) reported that countries with low income and high poverty levels had a higher prevalence of chronic kidney disease and renal failure among their population than countries with low poverty levels, as evidenced by low screening and diagnosis of patients. Similar results to current research were also established in prior research conducted by Novick et al. (2022), who reported that persons facing homelessness were more likely to contract chronic kidney disease than people with stable housing as they lacked medical insurance coverage and access to kidney health services, thereby resulting in increased renal failure among such people.

Whereas current research highlighted that SES is a significant predictor of renal failure, previously demonstrated that hospitalization because chronic kidney disease diagnosis in low-income areas was high compared to high-income areas because of limited access to kidney health services and comprehensive medical insurance covers (Saunders et al., 2020). The prevalence of chronic kidney disease is directly correlated with poverty, which has contributed to a lack of adequate health services and medical insurance resulting in subsequent increased chronic kidney disease and renal failure.

RQ7: Is there a relationship between renal failure and the number of procedures?

Current research findings established that the number of procedures was a significant predictor of renal failure, and the increasing number of procedures results in an increased likelihood of renal failure. The results imply that the high number of procedures results in more renal failure due to increased kidney complications. Although current findings indicated that the number of procedures predicted renal failure, Schrauben et al. (2021) established that education levels affected the number of procedures through medication adherence, affecting renal failure among the patients. Education level determines the income, adherence to medical procedures, and access to medical insurance by patients in self-management of chronic kidney disease, thereby reducing renal failure. However, a lack of adherence to medication procedures may result in increased renal failure (El Shamy & Cavanaugh, 2020). The findings have contributed to past research by revealing that several procedures significantly predict renal failure among patients suffering from chronic kidney disease.

RQ8: Is there a relationship between renal failure and the number of diagnoses?

The number of diagnoses was a significant predictor of renal failure, and an increase in the number of diagnoses increases the likelihood of renal failure. The implication is that the number of kidney disease diagnoses may predict the patient's likelihood of experiencing renal failure. The results concur with prior research findings, which highlighted that individuals suffering from chronic kidney disease but with prior knowledge of the same managed the adverse effects of chronic kidney disease compared to individuals with limited renal disease information, thereby reducing the number of diagnoses and renal failure (Shah et al., 2021). Consistent with current findings, Chu et al. (2020) reported that individuals with limited health knowledge were at a higher risk of developing chronic kidney disease and an increased number of diagnoses than health-literate individuals. The prevalence of chronic kidney disease and renal failure is because of an increased number of diagnoses which is high among patients with low literacy levels (Chu et al., 2020).

RQ9: Is there a relationship between renal failure and comorbidities?

The logistic regression results revealed that several comorbidities were a significant predictor of renal failure, such that increasing comorbidities increase the likelihood of renal failure. The findings indicate that the high number of comorbidities results from an increased likelihood of renal failure among patients with chronic kidney diseases. While the finding above indicates that comorbidities predict renal failure, previous research demonstrated that CKD is associated with comorbidities that increase its severity and burden (Jacob et al., 2019). Hypertensive nephropathy and diabetic nephropathy were the leading causes of CKD. Further, retinopathy, diabetes, hypertension, cardiovascular disease, and chronic obstructive pulmonary disease were common CKD-associated comorbidities (Jacob et al., 2019). Sullivan et al. (2020) revealed that CKD development was increased in patients at risk of multimorbidity, including cardiovascular diseases that accelerated the progression of kidney failures, resulting in renal failure. However, MacRae et al. (2020) explained that being the leading cause of mortality and morbidity globally, CKD is common to people with underlying medical conditions such as diabetes, hypertension, and cardiovascular disease due to deteriorated kidney functions, contributing to renal failure.

RQ10: Do length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and comorbidities collectively predict renal failure?

The findings established that length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and comorbidities collectively predict renal failure. The findings imply that renal failure among patients with chronic kidney diseases may be associated with length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, diagnoses, and comorbidities. The findings agree with previous literature. Similar to the current findings above, past research revealed that length of stay, totally charged, age, and gender are significant predictors of renal failure (Tong et al., 2022; Soldati et al., 2022; Morton & Shah, 2020).

The findings also concurred with past literature, which indicated that SES, number of procedures, number of diagnoses, and comorbidities were significant predictors of renal failure among patients suffering from chronic kidney diseases (Sullivan et al., 2020; Novick et al., 2022; Schrauben et al., 2021; Chu et al., 2020. The findings have added to past literature by establishing that the length of stay, total charges or cost, age of the patient, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities collectively predict renal failure among patients suffering from chronic kidney diseases.

The multivariable results (Table 17; Figure 22) add an important qualification to the interpretation of RQ1 and RQ7. Once the other predictors were held constant, length of stay was no longer positively associated with renal failure, and the number of procedures was no longer statistically significant. The unadjusted associations for these two hospitalization characteristics therefore appear to reflect the greater age, diagnostic burden and comorbidity of patients with renal failure, rather than an independent effect.

Limitations of the study

One limitation of this research was that the researcher used a quantitative study design. One of the limitations of quantitative research design is that it focuses on numerical data and may not give the perception and opinions of participants regarding a particular topic of interest. The researcher also used secondary data, and secondary data sources may not provide primary first-hand information needed for the authenticity and reliability of study findings. Another limitation was that data were collected with a specific data boundary of three years which may not reflect the current status regarding chronic kidney diseases and renal failures. In addition, because renal failure was present in fewer than one in ten discharges, the overall classification rate of the multivariable model (90.3%) did not exceed the 90.5% that would be obtained by always predicting the absence of renal failure. Classification at the default 0.5 cut-off is therefore not informative for an outcome of this prevalence, and the area under the ROC curve (.856) is the more appropriate measure of the model's discriminative ability. Finally, the unit of analysis

was the hospital discharge rather than the individual patient, so patients admitted more than once during 2010–2012 may contribute more than one record.

Implications for practice

Health practitioners may use the study findings to slow the progression of renal failure by creating awareness regarding the need for early detection and treatment of kidney diseases. McQueen et al. (2017) emphasized the significance of early detection and clinical intervention for patients with diabetes who have CKD. Nephrologists may use these study findings to meet the needs of their patients because of a better understanding of the differences between sexes in the etiology, mechanisms, and epidemiology of renal failure. To address disparities in renal failure patients' access to care and outcomes, Carrero et al. recommended addressing societal factors as well as variations in the underlying pathophysiology of the disease. Healthcare organizations may therefore use this study results to promote healthcare equality across different racial groups. According to Kim et al. (2018), doctors should concentrate on restoring renal function and closely monitoring patients with risk factors for developing CKD. Therefore, CKD patients, nephrologists, physicians, and the local communities could all benefit from this study's findings.

Recommendations for Future Research

The research was limited to a quantitative study design which may not give the perception and opinions of participants regarding a particular topic of interest. Future research should be conducted using a qualitative research design to provide the views and perceptions of participants regarding how the length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities predict renal failures. Secondary data sources may not provide primary first-hand information needed for the authenticity and reliability of study findings. Therefore, future research should consider using primary sources of information to enhance the reliability of the study results. Researchers should also use this

study's findings to investigate how socioeconomic factors predict renal failure and the causes of health disparities across different races.

Conclusion

The purpose of this study was to investigate the relationship between renal failure and length of stay, total charges, and their association with age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities in a cohort of inpatients in the United States. The results revealed that, considered individually, longer length of stay, higher total charges, older age, more procedures, more diagnoses, more comorbidities, male sex, and Black race (compared to White) were each associated with an increased likelihood of renal failure, while higher SES was associated with a decreased likelihood. Considered together, these variables collectively predicted renal failure with excellent discrimination; after mutual adjustment, age, sex, race, SES, number of diagnoses, and number of comorbidities remained independent predictors, whereas the associations for length of stay and number of procedures were explained largely by the other variables. The findings have contributed to the body of past literature by indicating the relationship between the length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses, and comorbidities and renal failure among patients suffering from chronic kidney diseases.

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